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Chapter 2 Crane maintenance instructions

Summary：

The service life of crane and auxiliary components greatly depends on the purchaser's **proper use, safe operation and timely maintenance**.

Chapter 2 of the book "STRUCTURE AND MAINTENANCE MANUAL for QUAYSIDE CONTAINER CRANES" gives a system expatiation on how to maintain correctly and timely. Crane maintenance mainly includes the following five parts: **cleanliness, fastening, adjustment, lubrication and inspection**. In order to fully achieve its potential service life, timely maintenance is very important for the crane and auxiliary components.

Once the auxiliary components reach their service life or limit, they should **be replaced** and this will be detailed in chapter 3. This book "*Quayside Container Crane Configuration and Maintenance Manual* " is mainly composed of the following six sections: **cleanliness, fastening, adjustment, lubrication, inspection and replacing.**

2.1. General

Crane Maintenance is composed of **Inspection, Routine Maintenance** and **Repair**。This Chapter basically expands upon the operating maintenance and routine inspection of the crane.

The inspection and routine maintenance is used to check whether the crane is in a safe state and to place the crane into a condition where a high confidence of reliable operation can be developed. The main work of the inspection and routine maintenance includes cleanliness, lubrication, fasteners and connections, spot-check and periodic inspection.

It is basically a visual inspection and some functional/detective tests. If, in doing so, an adequate assessment is not possible, a disassembly of crane parts has to be undertaken to get access to the hidden parts. Chapter 3 "The repair Instructions" will illustrate crane repair procedures in detail on components which need disassembly and disengagement.

The cautions and instructions about inspection and maintenance are outlined, aiming at giving a general idea on how to carry out an inspection and a routine maintenance.

Inspection of cleanliness
Inspection of fasteners and connections
Inspection of concentricity (alignment) periodically
Inspection of lubrication
Inspection and preventive actions

In addition to this maintenance manual, all subcontractors have manuals for their own components. To maintain such components, please refer to their manuals. The list of these manuals is in Table 2-1 as follows.

Table 2-1： Manual from Subcontractors（Mechanical Part）

Reducer	ZPMC
Motor Brakes	Bubenzer
Drum Brakes	Bubenzer
Wheel Brakes	ZPMC
Drum coupling	ZPMC
Motor coupling	ZPMC
Spreader	ZPMC
Overheight adapter	ZPMC
Elevator	ZPMC
Spreader cable reel	ZPMC
Power Cable Reel	ZPMC
Service crane	ZPMC
Wire rope	Tokyo
Air compressor	China product
Bufffer(gangtry)	ZPMC
Drum、Sheave、Wheel	ZPMC
Trolley power chain system	IGUS

2.2. Cautions and safety notes



For repairing parts or replacing components safely, maintenance persons must follow the description in this manual and manuals it has mentioned. The relevant sections of this manual are to be studied carefully before every maintenance or repair.

Before starting any maintenance or repair job, this section should be read thoroughly and carefully in order to get clear general ideas on maintenance safety and instructions. Safety is always a first priority in any maintenance operation. When a crane is under maintenance, some of its parts may be disassembled or unstable, thus special attention should be paid. Operators and maintenance personnel should be familiar with the crane's features. Before operating or maintaining the crane, maintenance and operation instructions have to be read carefully and understood by all maintenance and repair personnel. The crane should be put into operation only by authorized and trained specialists. Follow local safety laws and codes issued by the related authorities Safety requirements mentioned in this manual, as well as special safety cautions on the maintenance tasks are all valid.

	**Always read this manual thoroughly before operating, inspecting and maintaining the cranes.Before commencement of service work, the container crane must always be made free of load, set to maintenance position and in a LOCKOUT/TAGOUT condition on effected systems if no other note is given.**
	**Refrain from any working method that may inhibit safety!**
	**Danger of falling**
	**Danger of tripping!**
	**Keep hands and body clear of moving or rotating parts!**
	**Wear protective clothing!**
	**Hazardous voltage**
	**Keep away from suspended loads**

The following are the safety instructions in maintenance.

2.2.1 Safety rules of operation

1) Follow the safety laws and codes issued by local government. Safety requirements mentioned by the user, manufacturer and fittings supplier of auxiliary materials are all valid.

2) Only qualified, trained, and designated personnel may carry out the repairs and maintenance on the crane.

3) The crane should be set to maintenance mode and safety methods should be adopted for such tasks.

4) Use safety harnesses and other safety gears as needed when working where you do not have a safe working platform and handrails to protect you. It is a good practice to work as a team of at least two persons when engaging in maintenance and repair work where there is potential danger.

5) Where inspections or work is being performed during the mechanism motion or rotation, enough spotters must be present watching each worker and ready to depress an Emergency Stop. Radios should be used to maintain communications with the operator. When a person stands on the trolley during maintenance, he must wear safety harness, and install safety chains near the rope inspection platform. When the trolley requires moving during the maintenance or inspection, even at a very slow speed, the person on the trolley must stay within designated platforms/hand-railing and pay special attention to any potential crush hazard.

6) Secure the repair area by closing a wider area and set striking signs and symbols about safety warning to prevent other idle person from entering.

7) During maintenance on the crane, be sure to close hatches on the machinery house and other places, or to protect them with fences. Where inspections or work is being performed without designated platforms/hand-railing, safety harness and other safety equipment should be used.

2.2.2 Safety Rules During repairing or maintenance

1) If hands will be on rope for inspection, lubrication or for any reason, they must be rope traveling away from pinch point.

2) The maintenance persons should pay particular attention to the moving/rotating parts.

3) An additional person is needed to act as a spotter between the operator and the mechanic while inspecting the rope.

4) When handling parts from the hatches on the floor in the machinery house, install protection around the hatch with its fenders before opening it.

5) Conductive parts must be switched off and secured against reactivation.

6) Provide absorbing devices for leaking operating materials.

7) Wear protective clothing (oil and petrol-resistant gloves, protective clothing, eye goggle, helmet, protective footwear)

8) Handle operating and cleaning detergents and solvents cautiously.

9) Provide a safe and environmentally friendly release, storage and disposal of operating and auxiliary materials.

10) Do not use machine parts as climbing aids.

11) Reinstall and test safety devices (which have been disassembled during maintenance or test and repair work) immediately after the end of maintenance and repair work.

12) Inform the operating personnel before the commencement of special or maintenance work.

13) Tighten screw joints which were loosened during test and maintenance work correctly.

14) Replace fastening elements and seals (e.g. self-securing nuts, washers, split pins, O-rings and seals) which cannot be used again

2.2.3 Safety Rules after Maintenance / Repair Work

1) Update fault log and maintenance check list. The responsible maintenance personnel must make also assure that the occurred faults are logged and eliminated.

2) Put away tools, foreign parts and operating materials left lying around.

3) The cranes have been provided with non-slip grating walking surfaces where possible. Many of these walking surfaces are more than 40m high at the cab and machinery house elevation. An object, which falls from such a height to the ground, is traveling at high speeds that can cause serious injury or even death if it strikes a person on the ground. It is everyone's duty and

responsibility to pick up any object or debris found on the crane to avoid accidents.

4) Test running and functional check of the components and safety devices. A functional test has to be carried out before starting operation and service again.

2.3. Cleanliness

The dust, oil dirt, water, impurity interfusing into the lubricant, rust and etc. may erode, speedup abrasion and depress even destroy partial original performance like reduce the braking moment, bring the electrical system short circuit or other malfunction, infect the leak-proofness of the hydraulic system and even destroy the hydraulic parts and bring severe result like system malfunction. Cleanliness is not only to keep a friendly environment but also to keep the machine super performance and prolong its lifetime. So in the routine maintenance the impassive contamination to the parts and equipments should be timely removed and keep the equipment leak tightness, without dust and oil dirt. The following are the routine methods and typical examples to keep the crane clean.

2.3.1 Major methods of cleanliness

2.3.1.1. Dry cleaning

"Dry cleaning" used here means adopting physical methods to remove the contamination from the surface without water or solvent. Shown in figure 2-1, the main methods including:

1) The cleaner can directly remove the dust and small dirt from the body surface. To create a clean environment and improve the service life and stability of equipment, the place where mechanical and electric equipments concentrate like machinery house, electric house and cab should be cleaned periodly.

2) Air compressor and air blower can directly remove the powder, iron chipping and other waste falling off from body surface easily. Its application including: cleanness of the faying face of the precision components such as bearing and deadeye, and wiping off dust from the surface of complicated shape part, electric element and other elements .

3) Cleaning by machining:

   a. Use wire brush or electric polisher with copper brush. This method can remove the flotation rust or metal surface treatment before repairing paint.

   b. Cleanness by finish machining such as turning or grinding. This method only is adopted in renovating the defective surface such as bulge or sunken in the brake disc, brake lining and so on.

2.3.1.2. Using cleanser

The cleanser is needed when the oil dirt on the surface is difficult to remove. In choosing the cleanser, advert to the problem that the solvent or cleanser based on halogenated hydrocarbon like trichloroethylene and dichloromethane will bring chemical reaction (injecting little water into the trichloroethylene, hydrochloric acid will generate) with the aluminum part, austenitic steel and galvanized part, and the part will be oxygenized. On extreme condition, such reaction will bring explosion. So the solvent or cleanser based on halogenated hydrocarbon shouldn't be used when cleaning the metal parts of the crane. In addition, the acid and alkali liquor shouldn't be used to clean the mental parts of the crane either, or it will destroy the parts directly. To cope with the common dirt, such cleanser as general gasoline bought in the market and industrial benzene can be used. As for the dogged dirt, use the cleanser such as dimethyl benzene.



Industry Cleaner



Air Compressor and Air Blower





Wire Brush、Copper Brush and Corner Grinder

Figure 2-1：Routine dry cleaning method

⚠	**Attention: Necessary protective equipments such as the goggles, gauntlet and costume should be worn to prohibit hurting to the body by powder, scrap iron and even the cold wind.**
⚠	**Attention: Some cleansers such as methylbenzene, dimethyle benzene, acetone, and trichloroethylene are poisonous. The related instruction book should be read carefully before using it.**

2.3.2 Main cleaning work

The following are some typical examples of cleanliness：

2.3.2.1 CLEANLINESS OF COMMON PARTS

1) Braking disc (braking wheel) and the brake lining (braking shoe) must be kept clean on the surface ***at all times***. No oil dirt, dust and rust, convex block, burr and painting coat will be permitted. Every rotating pin shaft of brake should be kept clean and free in rotation. No paint contamination and rust will be permitted (shown as Figure 2-2).



Figure 2-2：Oil dirt and rust on the brake disc must be removed

2) The cooling fan filter of the motor should be kept clean and properly ventilated (shown as Figure 2-4). As to the DC motor, the commutator should be kept clean.



Figure 2-3：Fan and fan filter

3) The internal of gear reducers should be kept clean. The bottom of the housing must be cleaned when oil is replaced.

4) The trolley running rail surface as well as both sides of rail should be clean。

5) The gantry running rail surface as well as both sides of rail should be clean, and no foreign matter is permitted.

6) The elevator guiding rail should be clean. No foreign matter is permitted.

7) The position easy to be corroded by seawater should be cleaned periodically.

8) All glass in the operator's cab, main control room and boom control room should be cleaned frequently to ensure a good view for the operators (as Figure 2-5).

9) Indication lights should be cleaned if there are powder dust and other contamination to ensure that the operators and maintenance personnel can identify colorful lights clearly.

10) All rear mirrors should be kept clean.



Figure 2-5： Operator's cab

11) Remove excess grease and oil resulting from lubrication of the crane components, both as part of the lubricating activity and following a short period of operation.

12) The parts should be cleaned before reassembling.

2.3.2.2 Cleanliness of the electric elements and system

1) Powder dust should be cleaned periodically for the high-voltage transformer to avoid leakage.

2) The electric control room should be kept clean. Doors and windows should be closed, and powder dust inside control panels and other places should be cleaned by vacuum cleaner. Pressure fan should be operated at all times.

3) Using dust collector to clean the place easy to sorb dust like the controlling panels and computer screens in the electric and CMS house (as figure Figure 2-6).

Figure 2-4：Fan and fan filter



Figure 2-6：PLC controlling panel

2.3.2.3 Cleanliness of the hydraulic system

Clean oil is the life of the hydraulic system. If impurity trespass the small damping holes or the gap in the precision component, the component will be pulled damaged and the oil roads will be jammed, which will bring danger to the safety running of the hydraulic system. Generally the impurity inbreak methods include: unclean hydraulic oil, unclean oil filling tools, carelessness in filling oil and maintenance, impurity slipping from hydraulic elements and air or water entered into the hydraulic system.

A. The way to prohibit the impurity from inbreak

1) Before fill oil, the hydraulic oil should be filtered, the oil filling tools should be clean and the filter placed at the entry of the oil tank should not to be removed in order to speed up filling. To prohibit the solid or fibre impurity from inbreak, the oiler should keep the glove and clothing clean.

2) In maintenance when the oil filling cover, the filter cover, inspecting holes or the oil pipe is taken down, keep way from the flying dust. The cover of the oil tank can not be opened until the dirt around the tank cover has been removed and the impurity at the connection place has also been removed after loosing the bolts (to refrain the water from infiltration, the water can not be used to clean the impurity). Choose the cleaning materials without fibre impurity dropping off and hammer with rubber beating surface if they have to be used. Hydraulic elements and hoses should be cleaned and blow dry before assembling. The filter also should be cleaned when oil changes. The type of cleaning oil should be the same as that of the system, the oil temperature should be kept between 45～80℃. In order to completely remove the impurity in the system, keep cleaning oil flow in large capacity. Hydraulic system should be cleaned over three times and after each cleaning the cleaning oil used should be sent out completely before it cool down. Before—fill new oil, the filter should be cleaned and new filter element should be adopted.

B. Keep the air out of the hydraulic system

The entrained air in the hydraulics will cause elements cavitation erosion, make the compressibility increased and the operation instability, reduce the work efficiency and make the actuator “crawl”. In addition, the air will also make the oil oxidation and speeding up its metamorphism. To keep the air out of the hydraulics the following list should be paid attention to:

1) Random check should be conducted after maintenance and changing oil, and then begin normal work. As shown in the Figure Figure 2-7 is the exhaust valve of the snag –protection cylinder used in the girder.

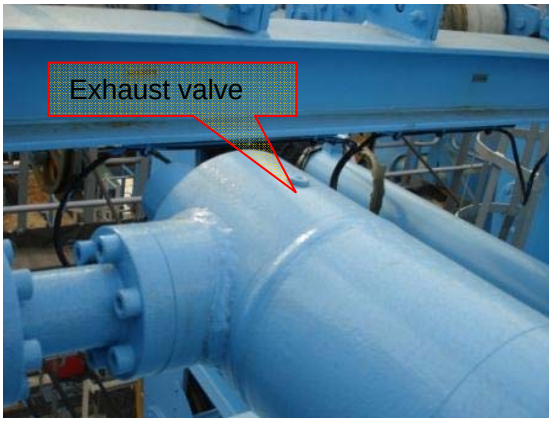


Figure 2-7：Exhaust valve of the snag –protection cylinder used in the girder.

2) The mouth of the inlet pipe in the pump can not come out of the oil lever and keep the inlet pipe being well sealed.

3) To keep well obturation for the driving axis of the lubropump, oil seal certified commended by the supplier should be adopted and ensure the function of airproof and oil proofing.

C. Prevent water penetration into hydraulics

If the oil contains excessive water, that will make the hydraulic elements corroded, the oil emulsified and metamorphosed, the tenacity of lubricating film reduced and abrasion speedup up. If the water in the oil is super standard, the hydraulics needs to be clean thoroughly and the oil should to be replaced.

D. Inspection of the filter element

Filter is the cleaning equipment used in the hydraulic system. As shown in figure Figure 2-8 is the filter of the hydraulic power package located on the girder. Inspect the squalidity degree of the filter and the characters of the impurity on the filter element to provide evidence for further fault diagnosis.



Figure 2-8： Filter of the hydraulic power package located on the girder

2.4. Fastening

2.4.1 Grade of fasteners

Although all bolts used on the crane are based on Chinese standard, the Chinese standard for bolts is same as the ISO standard. The bolts are also equal to the American metric bolts in specification and have parallelism relation in grade. So it is very easy to find suitable bolts for replacement. TABLE 2-2 is another comparison table for bolt grade in different standards. The strength grade is usually marked in the head and end of the bolt (shown in Figure 2-9). Bolts used in the crane are usually greater than or equal to grade 8.8, so less bolts can not to be used in the maintenances.

It must to be noted that the grade classification as mentioned above is suited for the bolts with the material carbon or alloy steel, and it is different from the stainless-steel bolts. Detail information please refers to other related data. Generally stainless-steel bolts with grade A2-70, nuts and washers are used when the bolts are less than M14 in the crane, so the same grade bolts shall be used in maintenances.

Table 2-2： SAE Grade and ISO Grade

ISO898-1	SAE J429	ISO898-2	SAE J995
Bolt	Bolt	Nut	Nut
Grade 4.8	Grade 1	4	
Grade 5.8	Grade 2	5	Grade 2
Grade 6.8		6	
Grade 8.8	Grade 5	8	Grade 5
Grade 9.8		9	
Grade 10.9	Grade 8	10	Grade 8

Fasteners used for metal structures should be ISO Grade 8.8 and 10.9 (ASTM A325 or ASTM A490) high-strength bolts. The bolt head and nut are usually bigger than the common ones and the matched bolt, nut and two washers are supplied as a complete set (shown in Figure 2-9, a set of bolt connection with grade 10.9). There are firm requirements on the bolt application, and only some simple knowledge is offered in this chapter, for more detail information please refer to the related criterion. As for the bolts ASTM A325 and ASTM A490，the《Specification for structural joints using ASTM A325 or A490 bolts 》written by AISC can be referred to. Pretightening torque is required for many bolts and it is realized by using a calibrated torque wrench. As listed in Figure 2-10 are some typical torque wrenches.

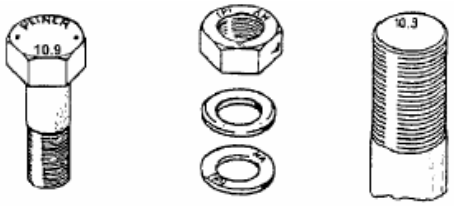


Figure 2-9：10.9 Grade fastener for structure



Hand torque wrench



Hydraulic torque wrench



Electric torque wrench

Figure 2-10：Typical torque wrench

2.4.2 Inspection of bolts

For cyclic loading in working the bolts may be loosened. In order to reduce stress amplitude, the bolts shall be pretightened to set value. Periodic inspection is aimed to insure the bolts loaded uniformly. Replace the bolts having yield and been lengthened or the deficiency-worried ones. Specification should be carried out during the downtime of the crane. There are two methods about the fastener specification：visual check and using wrest wrench，and visual check is mostly used

2.4.2.1. Visual check

Using the following methods to visually checking, refer to the Figure 2-11:

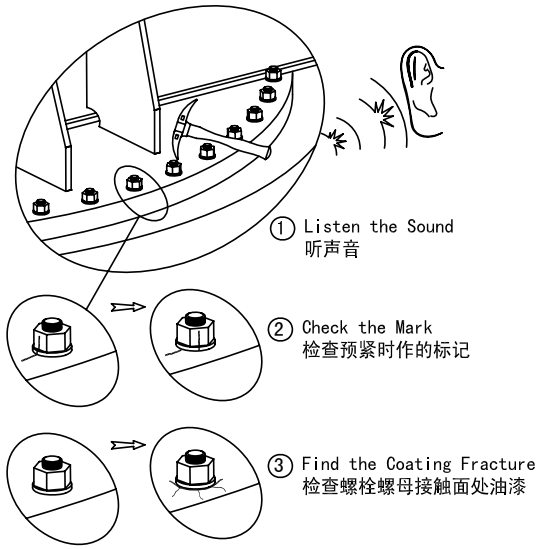


Figure 2-11：Methods of Finding a Loosened Bolt

1) The sound of a pointed hammer hitting the head of a bolt can indicate the looseness/tightness of the bolt.

2) The position between the nut and the connecting part should be visually checked (during the fabrication in the ZPMC's facility, the positions between some critical nuts and connecting parts will be marked out with a marking pen). If the marking positions are deviated, it indicates that the bolts are loose.

3) The paint film of nuts and connecting parts can also be visually inspected. If the paint film has fracture, the bolts should be inspected.

2.4.2.2. Use the torque spanner

A. Common fasteners（excluding the Fasteners used for metal structures）

1) Before inspection, the crane should be off working and make the bolts idly without working stress.

2) If needed, remove all the dirt and paint on the fasteners. Spray a thin layer of oil-lubricant rust remover (CRC or other equate product) on the nut and washer. The lubricant rust remover is functioned after 10~20 minutes.

3) Mark the original location by a marking pen on the nut and on the joint surface of the structure and the washer (shown at Figure 2-12).

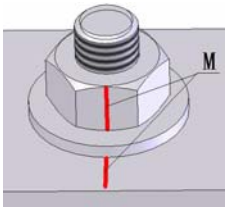


Figure 2-12：Marked bolt

4) An initial tension of 75% of the final tension shall be developed. Then the final tension shall be developed.

5) Check the nut: if the nut can be turned, make it out by a marking pen or painting and refasten or replace it later on.

6) If needed, the bolt joint should be replaced.

7) Repeat the process 3)-6) to inspect the other bolt joints.

8) The rotating nut should be refastened after the loosing the bolt and according to the methods mentioned in the section 2.4.3.3-A.

9) The non-rotating nut should be refastened according to 10) ~11) step by step.

10) Turn the nut about 30°by a torque spanner to loosen the nut.

11) Refasten the nut to its pretightening torque requirement.

12) Remove all the dirt and use the proper solvent to clean bolts, nuts, washer and places around the joint surface of the structure, and then repair the damaged paints according to paint repairing process.

B. Fasteners structures（**take the flanged connection as an example**）

1) Before inspection, make the crane under stopped state and make the bolts idly without working stress (without loading).

2) If needed, remove all the dirt and paint on the

fasteners. Spray a thin layer of oil-lubricant rust remover (CRC or other equate product) on the nut and washer. The lubricant rust remover is functioned after 10~20 minutes.

3) Use a marking pen to mark out the original position on the nut and the surface between the connection part and the washer (shown in the Figure 2-12).

~~4)~~ An initial tension of 75% of the final tension shall be developed by a marked torque wrench.

5) Check the nut: if the nut can be turned, make it out by a marking pen or painting and refasten or replace it later on.

6) Repeat the process 3)-6) to inspect the other bolt joints.

7) Account all the nuts that marked rotation and should be replaced.

8) If the account of the bolts marked to be replaced exceeds 20%, use the same grade bolts to replace them according to the process mentioned in the section 2.4.3.3-B.

9) If the account of the bolts marked to be replaced under 20%, replace all the marked bolts and the bolts on both sides, and refer to the process in the section 2.4.3.3-B.

10) The non-rotating nuts should be refastened according to 11) ~13) step by step.

11) Turn the nut about 30°by a torque spanner to loosen the nut.

12) Refasten the nut to its pretightening torque requirement.

13) Use proper solvent to clean the bolt, nut, washer and the joint surface of the structure. Then repair the paint according to the "paint repair process" mentioned in the appendix A.

2.4.2.3. Inspection period for the fasteners

Table 2-3：Inspection Period for the Fasteners

Inspection Type	Inspection Period （before the first maintenance）*	Inspection Period （after the first maintenance）*
Visually check	Per 3 months （except special sites）	Per 6 months （except special sites）
Use torque spanner	10% spot-checked (the sixth month or 50000th hour in work circulation)	the 5th, 15th、25th、30th、35th year
Replace fasteners	If needed	If needed

* The first maintenance here gives priority to the first expiration of the 6th month and 50000th hour in work circulation.

After the crane is put into operation for six months or 50000 hours work circulations, the tightness of bolts should be checked according to Table 2-3：. Because the bolts are easy to be loosened in the primary stage, so an all-around inspection should be carried out. High-strength bolts should be spot-checked by 10% for each group. If any one bolt is found loose, all bolts within this group should be thoroughly checked. In addition, the bolts bearing vibrating or varying load and the effective anti-loosening methods have not been adopted, the inspection period should be shortened to avoid accidents. Electrical wiring and connecting bolts of small electrical parts, such as bar wiring connection in control panels, motor wiring, junction box wiring, various kinds of limit switches, encoders and cam limit switches (inside and outside), usually have smaller area and size. Impact and vibration during the operation of the crane may cause the bolts to loosen. Once the bolts are loose, the electrical control will fail. Therefore, the inspection of fastening condition for this kind of bolts must be carried out, and the wiring condition should be inspected periodically.

2.4.3 Process for bolt's replacement and tightening

2.4.3.1. Replacement standard

The bolts under the following condition should be replaced：

1) The bolts required to be replaced mentioned in the above section.

2) The bolts with such defects like impairment, distortion, slide tooth, lacking tooth, corrosion, big change of the screw roughness and so on.

3) If the bolts following out the standard GB1228、GB1229、GB1230 ASTM325 and ASTM490 are hot dipped galvanizing, they are not allowed to be used repeatedly, and should be replaced once loosened after being final tightened. According to the standard AISC only the bolts by black finished can be used again with engineer's permission. In addition, when the adjacent bolts are loosened by tightening bolts, they are not

needed to be replaced.

2.4.3.2. Tightening torque for the bolts

As listed in the table 2-5 is the tightening torque for the bolts, studs and screws. if no special requirement, the data in the table can be used directly. Pretightening torque Mo = K*Fo*d (K is the torque factor, Fo is the pretightening force and d is the bolt diameter). The friction factor of the nut and washer, the nut and bolt changes will cause the torque factor change correspondingly, so even when the required pretightening force is unchanged, but the torque factor is changing. Generally under dry connection, the torque factor K=0.145, if a little lubricant or MoS_2 is filled, K=0.12. For the important connection, the torque factor is got by spot-check of the each bolt or that of the same connection and measuring through experiments. Before finishing tightening the fact torque factor of all the high-strength bolts or other similar connection should be measured.

Table 2-4：Tightening Torque of Stainless Steel Bolts with Small Size (Grade A2-70 级)

Size	M5	M6	M8	M10	M12	M14
Tightening torque (Nm)	3.2	5.5	13	26	46	74

Table 2-5：Tightening Torque of Major Bolts

Note:

* = without lubrication

** = with a little lubricant or MoS_2 lubrication

Size	Tightening torque (Nm)				Tightening force (N)	
	8.8s*	Grade 8.8s**	10.9s*	10.9s**	8.8s	10.9s
M 12	66	54	92	76	37766	53109
M 14	105	87	147	122	51520	72450
M 16	163	135	229	190	70336	98910
M 18	225	186	316	261	86016	120960
M 20	318	263	448	370	109760	154350
M 22	433	358	609	504	135744	190890
M 24	550	455	774	640	158144	222390
M 27	805	666	1132	937	205632	289170
M 30	1093	905	1537	1272	251328	353430
M 33	1488	1231	2092	1731	310912	437220
M 36	1911	1581	2687	2224	366016	514710
M 39	2473	2046	3477	2878	437248	614880
M 42	2851	2360	4009	3318	468168	658362
M 45	3818	3160	5369	4443	585109	822809
M 48	4593	3801	6459	5346	659973	928087
M 52	5937	4913	8349	6909	787401	1107282

The torque listed in the table is only suit for the connection surface without special requirements. The bolts in such connection are only bearing pretightening force, and the working force is vertical to the bolts and transferred through the friction from the connection surface. For those bolts bearing not only the pretightening force but also the working force, for example the bolts used in the bearing pedestal and the motor base, the pretightening force should be recalculated (the tightening torque is often less than the one listed in the table), please refer to the related drawing.

2.4.3.3. Bolt fastening and replacing process

A. Fastening process of the general blots (excluding the high strength bolts for structure) during the first installation or replacement.

1) Ensure the washer be planished and clean without oil dirt and burr。

2) Clean the connecting surface of the structure and washer and ensure the surface be smooth.

3) Clean the bolts and nuts.

4) Set well the washer and ensure the inclined plane touch the bolt head or the nut.

5) An initial tension of 75% of the final tension shall be developed. To ensure the flange bolts load equally, fasten the blots in the order from the rigid joint to free end or form the center of the bolt group to around. If the connection steel plate warps and the connecting bolts are fastened from the two ends to the middle, the middle of splice plate may plumps up and can not keep closely which will make friction forces lose partly. For detailed fastening process refer to Figure 2-13. Make marks on the large hexagon head and high strength bolts to identify their first fastening to avoid missing.

6) Finally fasten the blots to 100% required torque. The fastening order is similar to the first one. Generally use two kinds of color to distinguish the first and final fastening.

B. Replacement of the high strength bolts for structure（flange connection）

1) To ensure the same torque factor, the bolts replaced one time should be the same batch products.

2) Ensure the bolts be enough in quantity, have the correct strength degree and tightening torque. Revise the torque wrench repetitiously before using.

3) Ensure the washer be planished and clean without oil dirt and burr。

4) Clean the joint surface of the structure and washer and ensure the surface be smooth.

5) Clean the bolts and nuts.

6) Use a new bolt to replace the old one which marked to be replaced and an initial tension of 75% of the final tension shall be developed immediately. Ensure the blots to be replaced and fastened one by one.

7) The old bolts being taken down should be destroyed purposely to prevent from using again.

8) After all the bolts being replaced, the final tension shall be taken to all the new bolts in the order shown in Figure 2-13.

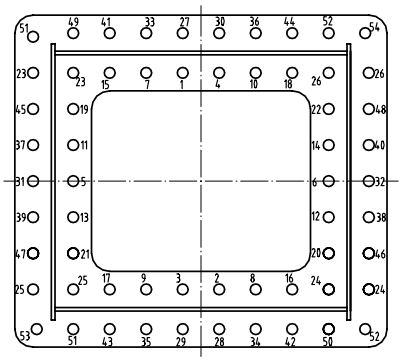


Figure 2-13：Fastening order of the flange bolts

2.4.4 Anti-loosing methods for bolts

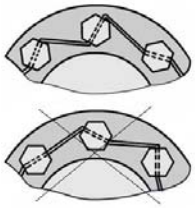
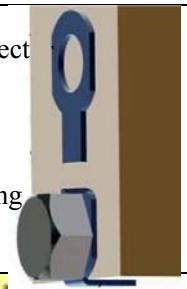
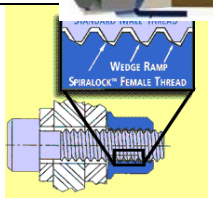
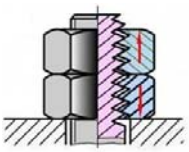
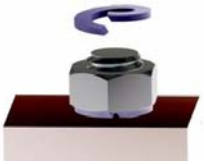
It is very important to adopt some methods to prevent bolts from loosing. If the bolts are loosened, the structure strength will be reduced and the downfallen bolts will bring danger to persons. Table 2-6：6 lists the main anti-loosening methods for the bolts used on the container crane.

2.4.5 Pin connection

Pins shall be well greased prior to each installation. After the installation, this process is to be repeated at all accessible spots of the pin.

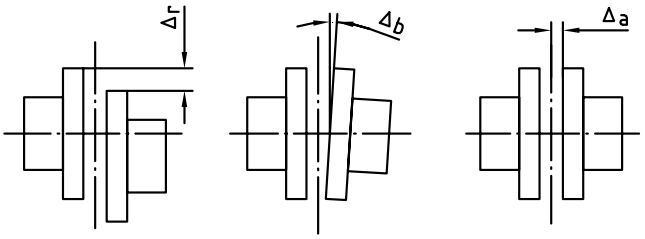
- If snag rings are used to secure pins, the snap rings must only be mounted once.
- New snap rings shall be used after a dismounting of the pins.

Table 2-6：Anti-loosening methods of the bolts used on the container cranE

METHODS	Instruction /application	Figure for example
using anti-loosing glue	The anti-loosing glue is often used in the fields with limited place, for example used on the fixing bolts of the anti-abrasion block for the wire rope.	
Lock washer and Fastening with wire rope	Both of these methods are comparatively reliable and have been widely used. Such methods are commended to be used in the fields without space restriction for example on the running or moving parts like braking disk, sheave and so on to prevent it from falling or flying off when the bolts are loosened.	 Correct Wrong 
Nylon anti-loosing nut SPL nut Double nuts	Nylon anti-loosing nut is the most widely used and it is reliable to anti-loose when matching with bolts. SPL nut has a strict requirement on the retightening torque and the nuts can be reused. Double nuts are not used commonly on the crane for inconvenience.	 
Spring washer or other anti-loosing washer	Spring washer is not very reliable and usually generates yellow water when used outdoors. Usually they are not used in ZPMC'S crane. But in some outsourcing or unimportant parts the stainless steel washers are still used.	 

2.5 Inspection of alignment

For rotary components such as couplings, inspection of alignment should be completed periodically. The alignment of couplings has great influence on the mechanism running and life. Therefore, the coupling alignment should be checked periodically. In the first year that the crane is brought into service, the inspection cycle should be shorter, e.g., once per three months. After the mechanism get stable in the service, the inspection cycle can be lengthened suitably. This inspection mainly contrapose the motor coupling and drum coupling of main hoist, boom hoist and trolley travel drive and the motor coupling of gantry travel drive. The standard of alignment is determined by the request of motor's alignment and the style of coupling. Figure 2-13 shows the main check items of coupling's alignment. Table 2-7 shows the standard's detail of coupling's alignment.



Radial misalignment Δr

Gap(angular displacement) Δb

Axial displacement Δa

Figure 2-13 The check items of coupling's alignment

Table 2-7： The standard's detail of coupling's alignment

Part	Style of coupling	Δr （mm）	Δb	Δa （mm）
Main hoist high-speed	Elastic coupling	≤0.05	0.12 mm	＝Δr

coupling				
Boom hoist high-speed coupling	Elastic coupling	≤0.05	0.12 mm	=Δr
Trolley high-speed coupling	Elastic coupling	≤0.05	0.12 mm	=Δr
Gantry high-speed coupling	Elastic coupling	≤0.05	0.06 mm	=Δr
Main hoist drum coupling	Gear coupling	---	10’	---
Boom hoist drum coupling	Gear coupling	---	10’	---
Trolley hoist drum coupling	Gear coupling	---	10’	---

2.5.1 Redress methods for alignment of motor and gear reducer

2.5.1.1 Use the laser alignment measuring instrument controlled by microcomputer.

Please see the following diagram and procedures.

1) Coupling needn’t be taken apart in the inspection. The laser generatrix is fixed on the shaft on one side of the coupling and the receiver is fixed on the other side. (See **Figure 2-14**). Both the generatrix and the receiver should round together.

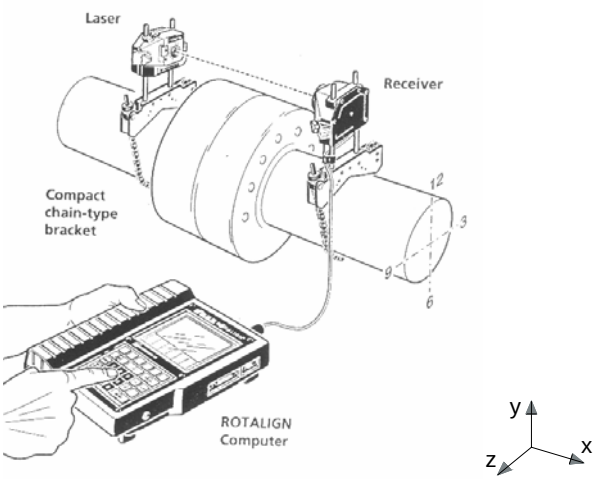


Figure 2-14 Alignment Check by Laser Measuring Instrument

2) Input correlative dimensions into the microcomputer, such as the dimensions of the motor pedestal, and the distance between the coupling center and the motor shaft ends (See **Figure 2-15**).

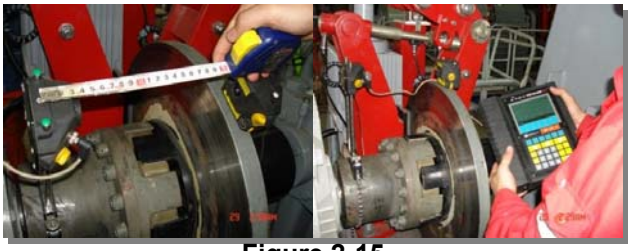


Figure 2-15

3) Before rotating, the initial value should be reset to zero.

4) Turning both the motor shaft and the reducer shaft only up to 60°-75°

5) Both the angular deviation of δ_z and the radial deviation of δ_y will show on the panel. The accuracy of the readings will be up to 1μm

6) The microcomputer will automatically calculate the value of displacement of the motor and the difference of the shim thickness which are advantageous for an operator to re-align the motor. The microcomputer has memory and printing function.

7) Add or reduce the shims to decrease the windage in elevation direction, adjust the fitting bolts or blocks for motor pedestal to decrease the windage in horizontal direction. See Figure 2-16.



Figure 2-16

8) Maybe the step 2~7 has to be repeated until the windages meet the standard.

2.5.2 Inspect the alignment with a dial indicator

Figure 2-17 shows the alignment measure method with a dial indicator.

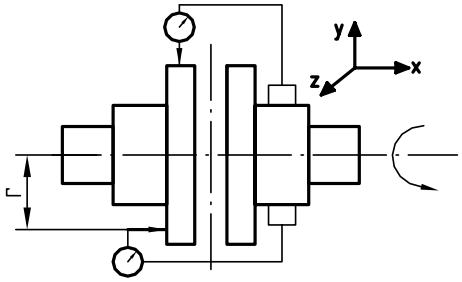


Figure 2-17 The measure method with a dial indicator

1) The indicator base is set on the shaft face of inner half-coupling and the finger contacts the other one.



FINGER 2-18

2) Initially, set the reading to zero. Suppose that the starting point is 0°, then the values of $\delta_{90°}$, $\delta_{180°}$ and $\delta_{270°}$ can be read after the coupling rounds 270° (please note the positive and negative signs). The circumferential jumping value in the vertical direction is $\Delta r_y = \delta_{180°} / 2$.

3) The circumferential jumping value along the horizontal central line is $\Delta r_z = (\delta_{90°} - \delta_{270°}) / 2$. The maximum circumferential jumping value is $\Delta r = \sqrt{\Delta r_y^{\,2} + \Delta r_z^{\,2}}$.

4) Similar as shown in the Figure 2-18, make the finger contact the end face of the other half-coupling or surface of the brake dick, and adjust the value to zero. After turning 270°, the values of $\delta_{90°}$, $\delta_{180°}$ and $\delta_{270°}$ can be read. And $\Delta a_y = \delta_{180°}$, $\Delta a_z = \delta_{90°} - \delta_{270°}$, then $\Delta b_y = arctg\Delta a_y / r$, $\Delta b_z = arctg\Delta a_z / r$.



FINGER 2-19

5) Estimate the thickness of shims under the motor pedestal and the translate in horizontal direction by Δr_x and Δr_y. And estimate circumrotation in direction Y (the windage of thickness between the front and back shims) as well as the circumrotation in direction X (the windage of thickness between the left and right shims) by Δa and Δb. Refer to 2.5.1.1 step 7) and adjust the estate of motor to meet the standard.

2.5.3 Adjustment methods of alignment for drum coupling

1) The indicator base is placed on one side of the coupling and the finger contacts the outer ring of the other side of the coupling. Refer to Figure 2-20.

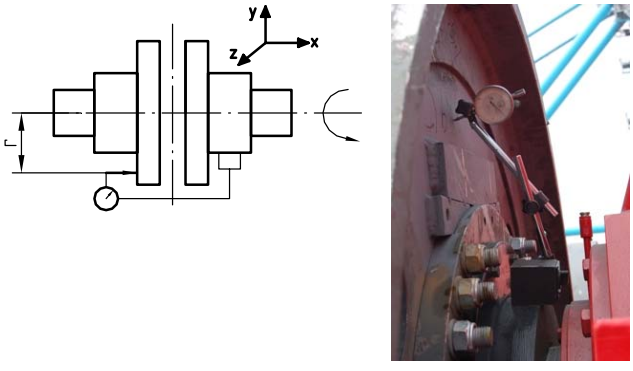


Figure 2-20：Inspect alignment of low-speed

coupling with dial indicator

2) Initially, set the reading to zero. So the values of $\delta_{90°}$, $\delta_{180°}$ and $\delta_{270°}$ can be read after turning 270° (please note the positive and negative signs).

3) Survey the radius r of measure-point.

4) Similar as shown in the Figure 2-18, make the finger contact the end surface of the other coupling or surface of the brake dick, and adjust the value to zero. After turning 270°, the values of $\delta_{90°}$, $\delta_{180°}$ and $\delta_{270°}$ can be read .And $\Delta a_y = \delta_{180°}$, $\Delta a_z = \delta_{90°} - \delta_{270°}$, then $\Delta b_y = arctg\Delta a_y / r$, $\Delta b_z = arctg\Delta a_z / r$.

5) Adjust the thickness of the shims under the drum bearing pedestal and the bearing pedestal's position to meet the standard.

6) The drums are supported by the low-speed shaft of reducer and in the center of it, so there are only angular deviation and axial deviation, no radial deviation.

2.5.4 Alignment of brake-disc

1) When the brake is open, the clearances between the two brake linings and disc should be the same. $\delta_1 = \delta_2 = 1 \sim 1.5$ mm. Refer to Figure 2-21

2) δ_1 and δ_2 should not be less than the minimum value of 1mm set for the brake. Refer to Figure 2-21.

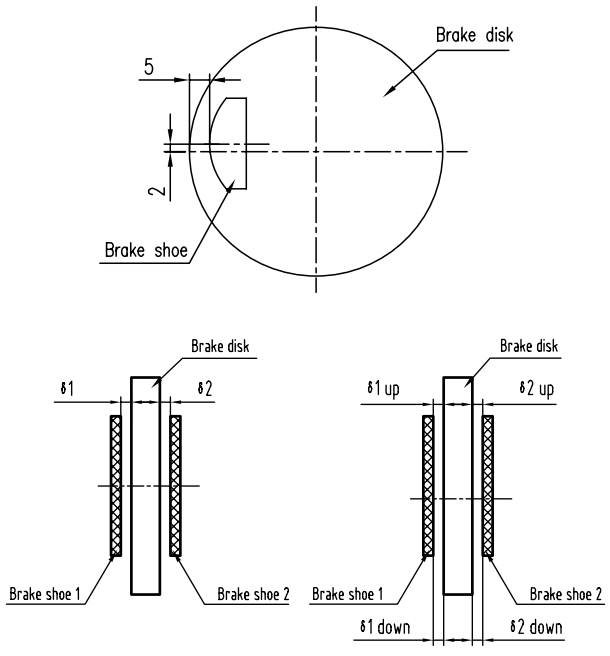




Figure 2-21： Clearance excursion of brake

3) The up and down clearance of brake disk should be the same, which means $\delta_{1\,up} = \delta_{1\,down}$, $\delta_{2\,up} = \delta_{2\,down}$. Refer to Figure 2-21.

4) Inspect the height of the brake to align its central line with the brake disc center. The deviation should not be more than 2mm. The horizontal deviation of the brake lining from the edge of disc should not be more than 5mm. Refer to Figure 2-21.

5) Inspect the position of the brake base in the motor axial direction. The deviation of the central line of disc thickness should not be more than 0.5mm. Refer to Figure 2-21

6) The contact area between the linings and the disc should not be less than 70% when brake is closed.

2.6 Lubrication

2.6.1 Summarization

Lubrication is a very important maintenance task. It will determine the normal service and endurance life of the components. Proper lubrication practice will help to avoid premature wear and extend the endurance life of the components. Lack of lubrication will destroy the components. For example, if an anti-friction bearing is short of lubrication, it will be damaged in a very short time. It is very important to monitor the temperature and sound level of the components to determine whether the components get enough lubrication. The lubrication of wire rope will be described in Chapter 4.

Attention: All used oil and grease must be collected and disposed properly!

2.6.2 Lubricant

For well effect of lubrication, the type of lubricants must be selected properly. Consistency should be maintained on lubricants. Do not mix up different lubricants together. If the lubricant supplier, type or grade is changed, launder the bearing, reducer, pin, and other devices thoroughly to remove the previous lubricant. If

different lubricants are mixed together, please contact the lubricants supplier.

The lubricants and grease used in the crane are listed in table 2-8~2-10. And the comparison of those products is also listed in the tables.

Table 2-8： Gear lubricants

	Oil volume L (/reducer)	Viscosity	Quantity (/crane)	Replaced Periodically
Main hoist	750	VG460	1	First time 500-800 hours Then every one year (mineral oil)
Boom hoist	450	VG460	1	
Trolley	150	VG320	1	First time 400 hours Then every 5000 hours or two years
Gantry	50	VG220	16	First time 400 hours Then every 5000 hours or two years
Emergency	28	VG220	2	First time 400 hours Then every 5000 hours or two years
Rope rereeving	9	VG220	2	First time 300 hours Then every 6 months
Elevator		N320	1	First time 40 hours Then every two years
Service crane	Hoist-1.7L，Trolley &Gantry-4.2L	Hoist-HD220,Trolley & Gantry-HD150	2	First time 150 hours Then every 1500 hours or one years
Gantry cable reel	36L	Shell spirax ASX 75W 90	2	First time 500 hours Then every 10000 hours
Spreader cable reel	5L	Shell spirax ASX 75W 90	2	First time 500 hours Then every 10000 hours

Table 2-9：Hydraulic lubricants

40°C viscosity(mm^2/s)		**Name and code name**	**°C pour point**	**°C Flash point**	**Place**	**Code name**
ISO VG 68	68	**Shell** Tellus HM	-24	223	Emergency pumping station, Back girder pumping station	B2

40°C viscosity(mm^2/s)		Name and code name	°C pour point	°C Flash point	Place	Code name
ISO VG 46	46	**Shell** Shell Tellus T Shell Tellus S Mobil DTE 20 Mobil DET 10M Mobil DTE Excel	-30 -39 -30 -27 -42 -33	218 246 218 232 216 226	Gantry wheel brake pumping station,	B3

Attention: Underlined trademark oil specially provided for ZPMC.

Table 2-10： Grease

NLGI Grade	Name and code name	drop point °C	Working point °C	Place	Code name
1 1	**Shell Alvania EP2** Mobil Mobilux EP 2	185 190	-20~+100 -25~+120	**Low-speed and high-loading sliding bearing: Equalizer pin, other pin, hinge point, etc.**	C1

Attention: Underlined trademark oil specially provided for ZPMC

2.6.3 Lubricant replacement standard

The deterioration of the lubricant lies on the work frequency and the ambience.

In inspection of the lubrication or in the use of lubrication, replace the lubricant whenever sizable quantities of any of the following conditions are found:

(1) Rubber-like deterioration

(2) Impurity

(3) Metal powder and badly discoloring

(4) Emulsification

(5) Separation of mineral oil and saponification radicel (deterioration due to oxidation).

In addition, sample specimen termly as follows to determine whether the lubricant should be replaced: take approximate 20 milliliters of lubricant from the machine and consult the lubricant manufacturer or deliver it to lab to check .The detail of standard is listed in TABLE 2-11, replace lubricant or append additives according to the test result if any standard is met. Refer to sampling program to Section 2.6.5.

Table 2-11：The standard of lubricant replacement

Test items	the standard of **replacement**	Test method
kinematics viscosity variety (40°C)	Increase 15% or decrease 20%	ASTM D 445
acid number	decrease 35% or increase ≥0.4mgKOH/g	ASTM D 974
normal heptane thawless	≥0.1%	IP 316
sheet copper erodibility	Less than Grade 2a	ASTM D 130
Chroma variety	Chroma ≥2 than new lubricant	ASTM D 1500
humidity	≥0.1%	ASTM D 1744

2.6.4 LUBRICATION PROCEDURE

(1) When the reducer oil is changed, drain the used oil while it is still warm. The tank floor must be cleaned out with oil to remove deposited silt, metallic particles and oil residue. Use the same type of oil to keep normal operation.

(2) When bearings are re-lubricated, pump the new grease into the oil pocket through grease fitting. Please clean out the old extruded grease after re-lubrication. During the first shift of operation after re-lubrication, the lubricating points should be cleaned.

(3) For lubricating motor, emergency brake, reducer and other outsourcing parts, please refer to their manuals.

(4) Wire ropes should be under good lubrication condition at all hours. Lubricating will prevent the wire rope from corrosion and reduce the friction in rope strands, between the wire ropes and the sheaves and/or drums. The type of lubricant and application frequency varies with the rope construction, the operating conditions and its functional application. For details, please refer to TABLE B-1 in Addenda B.

(5) The quantity of oil/grease should be suitable.

(6) For gear reducers (splash lubrication), the oil level should be kept a little higher than the central level between the highest and lowest levels because the oil level will become lower when the gear reducer is working.

(7) For bearings and other relatively sliding parts, it is natural that the old grease flows out of the seal from one side when the new grease is pumped in from other side.

(8) Oil and grease complementary cycle. It has relationship with the ambience, sealing condition, working grade and operating frequency of each part. For details, please refer to TABLE B-1 in addenda B.

(9) For reducers not used frequently, bottom valves should be opened termly to inspect the moisture content of the oil.

(10) At the early stage of the crane is put into operation, three samples should be collected from the replaced old oil to analyze the percentage of metal powder in it. Compare and analyze the lubricant to find out the working status of machine.

2.6.5 Lubrication Analyze and Monitoring

An effective oil conditioning and monitoring program can help extend the equipment life, reduce scheduled and unscheduled downtime, reduce maintenance costs and extend lubricant change intervals.

The condition of oil is very important for proper operation of a device. A Sample for each component in the table below should be taken at least once a year. An analysis of the sample should be carried out as a minimum:

1) Kinematical Viscosity

2) Particle Count

3) Moisture content inspected by Karl Fisher

4) Metal grains inspected by Spectroscopic

2.6.5.1 Basic principles of sampling of lubricant

1) Sample from the same point every time to the best of one's abilities.

2) Sample from the machine used latently and sample when the temperature of lubricant is near that at working time.

3) Sample from the lower reaches of working group. Refer to Figure 2-22. Don't sample from the points listed below:

Where there is only a little lubricant or lubricant is not ambulatory.
Where remnant and water conglomerating (for example, in the relief valve)
The end of blind-tube
The lower reaches of sieve valve
The sieve jaws on the bottom of the tank

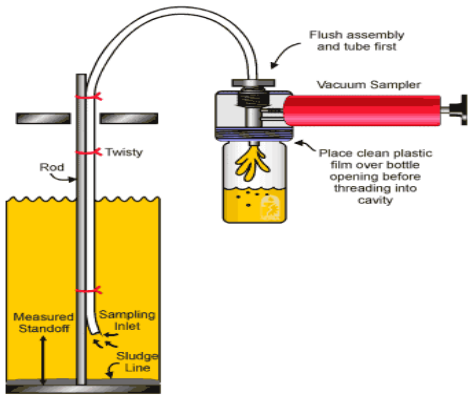


Figure 2-22： Sample points

4) Don't sample while the lubricant has just been changed.

5) Keep clean of the sampling tool (dry, no remnant).

2.6.5.2 Sampling tool in common use

Table 2-12： Sampling tool

	
sampling bottle	sampling canal
	
sampling pump	sample label / sampling point label

2.6.5.3 The steps of sampling

1) Take off the cover of a clean, dry sampling bottle and screw the bottle into the pump.

2) Shear the ends of a new section of pipe to 45 degree.

3) Tuck the pipe in the pump until the pipe into the bottle through the top of the pump, and then tweak the screw to fix the pipe.

4) Tuck the end of the pipe into the sampling point, keep the bottle upward. Absorb the lubricant to the bottle with the pump, and remain 25mm air clearance.

5) Turn off the bottle and tweak the cover to avoid pollution.

6) Fill in the sampling label clearly and stick it to the sampling bottle. The collection of lubricant is completed.



Figure 2-23: Step 1



Figure 2-24: Step 2



Figure 2-25: Step 3



Figure 2-26: Step 4



Figure 2-27: Step 6

2.6.6 Lubricating points layout

Figure 2-28 and figure 2-29 gives the assembly to be lubricated, the amount of lubricating points and the style of bearing at lubricating point. There are relationship between the figures and Table 2-13.

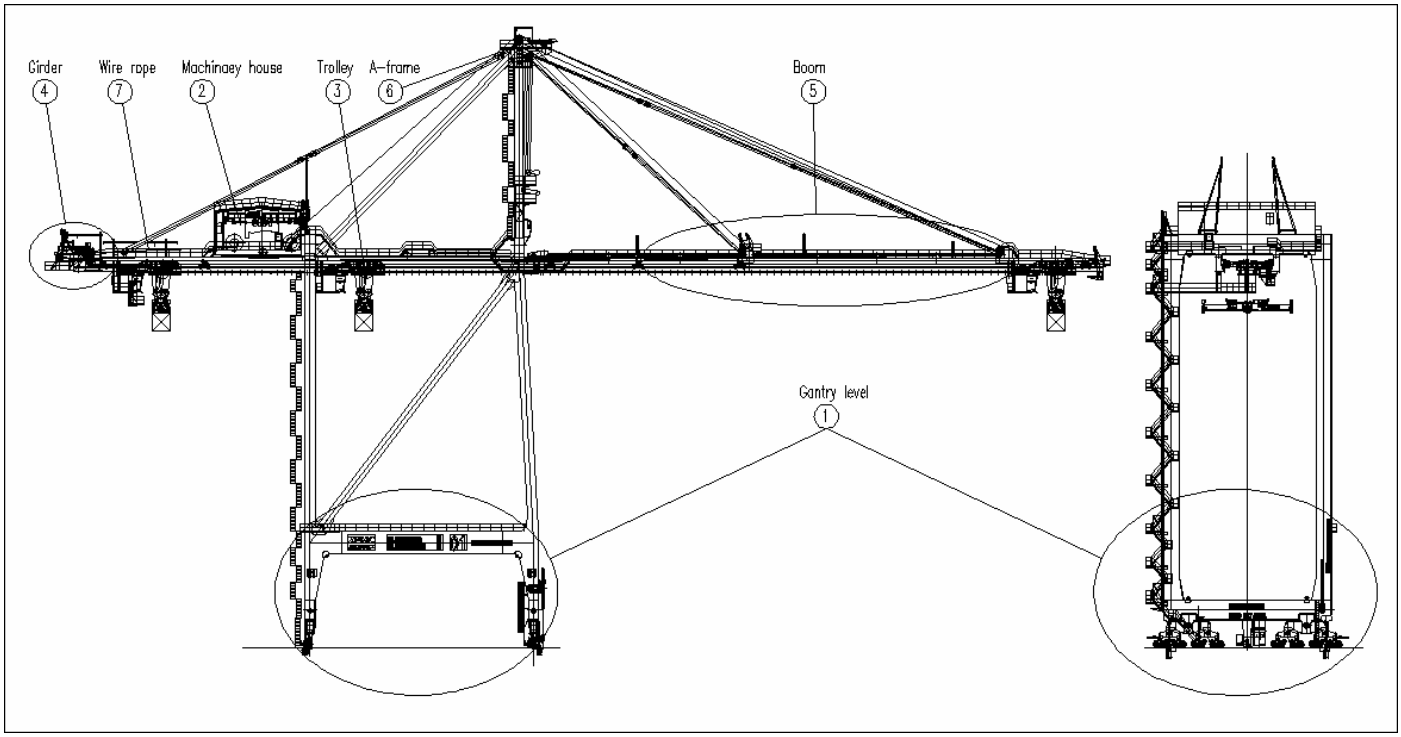


Figure 2-28 lubricating areas plot lubricating points layout

Lubrication point layout

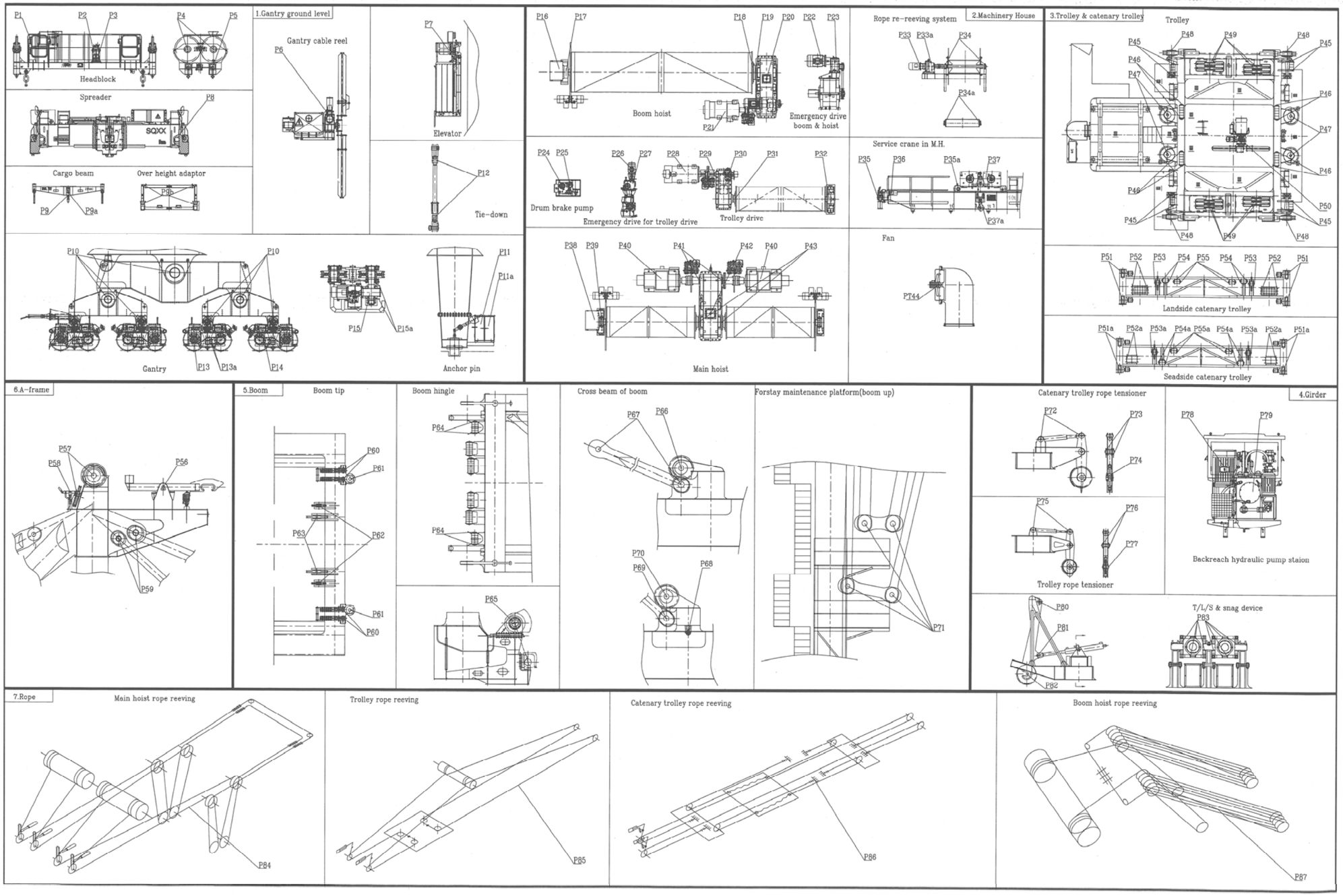


Figure 2-29 lubricating points layout

2.6.7 Lubricating frequency

The lubricants and lubricating frequency are listed in Table2-13. Lubricating frequency has relationship with the weather such as temperature and rainfall, and the service condition of the crane at the terminal. The lubricating frequency listed in the table can be used in common condition. The maintenance people should observe whether the lubricating is enough during the cranes service, to protract or shorten the lubricating periods properly. Refer to Figure 2-28: Lubricating areas plot and Figure 2-29 Lubricating points layout while using the table.



The lubricating frequency listed is based on ZPMC's experience and common condition. Please keep an intact note or log. Based on the maintenance note and well lubricating condition, the lubricating frequency can be redefined or changed



The interval of lubrication is at most 4 years commonly. After this period, the lubrication should be changed even if the lubrication is not metamorphosed. Refer to the lubrication suppliers' recommendation.



Sample the hydraulic oil and gear lubrication while the recommended replacing period listed in table 2-13 is reached. If the sample shows that the lubrication needn't be changed, the lubrication can be used while monitoring should be reinforced. What's more, gear lubrication should be sampled every 2 years even if the replacing period is not reached.



The lubrication of forestay is very important for keeping well working status. If there is rust on the friction surface between the pin and stay bar, appended bending moment will be engendered in the stay bar while it is turning, and flaw can be engendered and expanded. So please remove ancient grease and add new grease every month.

Table 2-13： Lubricating frequency

symbol			
D	Daily	3M	Every three month or～1000 hours
W	Weekly	6M	Every six month or～2000 hours
M	Monthly ～or 300 hours	A	Annually or～4000 hours
H	According to working time (hour) or else		
Abbreviation of lubrication （Refer toTable2-8～Table2-10）			
A1	VG 220	A2	VG320 320
B1	VG460	B2	Hydraulic oil
C1	Alvania EP2	C2	Wire rope lubrication oil.
C3	Refer to Motor manual		
Abbreviation Of Lubrication Points Types：			
Ⅰ	Hydraulic	Ⅱ	Gear
Ⅲ	Common anti-friction Bearing	Ⅳ	Motor Bearing
Ⅴ	Sliding Bearing	Ⅵ	Chain drive
Ⅶ	Gear Coupling or Spline	Ⅷ	Wire Rope
Ⅸ	Others such as spring, seal…		
🕮	Refer to drawing / correlative independent part maintenance manual		

Lubri-cation point P	period							lubricant (code name)	Lubrication point style / remark	amount	🕮
	D	W	M	3M	6M	A	H				
1			●					C1	Ⅴ/door hinge	4	
2			●					C1	Ⅴ/hinge	1	
3			●					C1	Ⅸ/spring	2	
4				●				C1	Ⅲ/hoist block bearing	4	
5			●					C1	Ⅴ/crank handle	4	
6								---	Gantry cable reel	-	Refer to Gantry cable reel maintenance manual
7								---	Elevator	-	Refer to Elevator maintenance manual
8								---	spreader		Refer to spreader maintenance manual
9				●				C1	Ⅴ/hook beam pin	2	
9a				●				C1	Ⅲ / hook beam thrust bearing	1	
9b				●				C1	Ⅴ/high tower rotate pin	-	Refer to over height adaptor maintenance manual
10				●				C1	Ⅴ/equalizer pin	56	
11								B1	Ⅰ /gantry wheel brake pump	2	Refer to hydraulic manual
11a						●		C3	Ⅳ/pump driving motor	4	Refer to motor specification
12			●					C1	Ⅸ/tie down	12	
13				●				C1	Ⅳ/wheel bearing	64	
13a				●				C1	Ⅴ/motor pedestal pin	32	
14						●		C3	Ⅳ/gantry driving motor	32	Refer to motor specification
15							●	A1	Ⅱ/gantry reducer：400h for the first time，then 5000h or two years	16	Refer to reducer maintenance manual
15a				●				C1	Ⅸ/reducer seal	32	
16						●		C1	Ⅱ/toothed clutch & cam limit reducer	1	
17					●			C1	Ⅲ / boom hoist drum bearing	1	
18					●			C1	Ⅶ/drum coupling	2	
19				●				C1	Ⅸ/reducer seal[1]	4	
20								A3	Ⅱ/ boom hoist reducer：500～800h for the first time，then one year	1	
21						●		C3	Ⅳ/ boom hoist driving motor	2	Refer to motor specification
22								C3	Ⅳ/ boom hoist emergency driving motor	2	Refer to motor specification
23								A1	Ⅱ/ boom hoist emergency reducer：400h for the first time，then 5000h or two years	1	Refer to reducer maintenance manual

Lubri-cation point P	period							lubricant (code name)	Lubrication point style / remark	amount	🕮
	D	W	M	3M	6M	A	H				
24						●		C3	Ⅳ/pump driving motor	2	Refer to motor specification
25						●		B1	Ⅰ/drum brake pump	1	
26						●		C3	Ⅳ /trolley emergency driving motor	2	Refer to motor specification
27								A1	Ⅱ / trolley emergency reducer：400h for the first time，then 5000h or two years	1	Refer to reducer maintenance manual
28						●		C3	Ⅳ/ trolley driving motor	1	Refer to motor specification
29								C1	Ⅸ/ [1] trolley reducer seal	3	
30								A2	Ⅱ / trolley driving reducer: 400h for the first time，then 5000h or two years	1	Refer to reducer maintenance manual
31					●			C1	Ⅶ/drum coupling	2	
32					●			C1	Ⅲ/ trolley drum bearing	1	
33					●			C3	Ⅳ / re-reeving device driving motor	2	Refer to motor specification
33a							●	A1	Ⅱ / re-reeving device reducer，replacing per 4 years	2	Refer to reducer maintenance manual
34					●			C1	Ⅴ /re-reeving drum sliding bearing	4	
34a					●			C1	Ⅲ/re-reeving rope roller bearing	8	
35								C1	Ⅲ/ traveling crane gantry wheel bearing		Refer to service crane maintenance manual
35a								C1	Ⅲ/ traveling crane trolley wheel bearing		
36								A1	Ⅱ /maintain traveling crane gantry / trolley reducer		
37								C3	Ⅳ/ gantry / trolley driving motor		
37a								C2	Ⅷ/ maintenance traveling crane wire		
38						●		C1	Ⅱ/toothed clutch & cam limit reducer	1	
39					●			C1	Ⅲ/ hoist drum bearing	2	
40						●		C3	Ⅳ/ hoist driving motor	4	Refer to motor specification
41				●				C1	Ⅸ/ [1] hoist reducer seal	4	
42								A3	Ⅱ/：hoist reducer: 500～800h for the first time，then every one year	1	Refer to reducer maintenance manual
43					●			C1	Ⅶ/drum coupling	4	
44								C3	Ⅳ/ fan motor	2	Refer to fan specification

Lubri-cation point P	period							lubricant (code name)	Lubrication point style / remark	amount	🕮
	D	W	M	3M	6M	A	H				
45					●			C1	III/hoist rope rollers	8	
46					●			C1	III/trolley rope rollers	8	
47				●				C1	Ⅴ/ trolley towed sheave bearing	4	
48			●					C1	Ⅲ/ main trolley wheel bearing	4	
49			●					C1	Ⅲ/ hoist sheave bearing	8	
50								---	Spreader cable reel	1	Refer to spreader cable reel manual
51				●				C1	Ⅲ/ carenary trolley wheel bearing	4	
51a				●				C1	Ⅲ/ carenary trolley wheel bearing	4	
52					●			C1	Ⅲ/ support roller bearing	8	
52a					●			C1	Ⅲ/ support roller bearing	8	
53				●				C1	Ⅲ/ sheave bearing	2	
53a				●				C1	Ⅲ/ sheave bearing	2	
54					●			C1	Ⅲ/ support roller bearing	8	
54a					●			C1	Ⅲ/ support roller bearing	8	
55					●			C1	Ⅲ/ support roller bearing	8	
55a					●			C1	Ⅲ/ support roller bearing	8	
56				●				C1	Ⅴ/ boom latch pin shaft	2	
57					●			C1	Ⅲ/ boom sheave bearing	14	
58					●			C1	Ⅴ/ rope slack pin shaft	2	
59					●			C1	Ⅴ/ pull rod pin shaft	4	
60			●					C1	V/ Load cell pin	4	
61						●		C1	Ⅴ/ guide sheave	4	
62			●					C1	III/catenary sheave	2	
63			●					C1	III/trolley sheave	2	
64			●					C1	Ⅴ /hoist rope roller bearing	4	
65			●					C1	Ⅴ/ boom hingle sliding bearing	4	
66					●			C1	Ⅲ/ boom sheave bearing	8	
67			●					C1	Ⅴ/ pull rod pin shaft	4	
68				●				C1	Ⅲ /trolley rope roller bearing	4	
69			●					C1	Ⅴ/ pull rod pin shaft	2	
70					●			C1	Ⅲ/ hoist sheave bearing	2	
71			●					C1	Ⅴ/ pull rod pin shaft	8	
72			●					C1	V/cylinder ball bearing	4	
73			●					C1	V/rocker pin shaft	4	
74			●					C1	III/catenary sheave bearing	2	
75			●					C1	V/cylinder ball bearing	4	
76			●					C1	V/rocker pin shaft	4	
77			●					C1	III/trolley sheave bearing	2	
78						●		C3	IV/pump drive motor	2	Refer to motor specifications
79						●		B1	I/backreach	1	Refer to hydraulic

Lubri-cation point P	period							lubricant (code name)	Lubrication point style / remark	amount	🕮
	D	W	M	3M	6M	A	H				
									multifunctional pump		manual
80			●					C1	V/pin shaft	4	
81			●					C1	V/cylinder ball bearing	4	
82			●					C1	III/Hoist sheave bearing	4	
83			●					C1	V/pin shaft	16	
84			●					C2	Ⅷ / hoist wire	2	
85			●					C2	Ⅷ / trolley wire	2	
86			●					C2	Ⅷ / catenary trolley rope	2	
87			●					C2	Ⅷ / boom wire	1	

Attention: To oil seal lubrication, add grease according to the direction of nameplate near the oil fitting, to avoid redundant grease come into the tank and mix with the gear lubrication.

2.7 Prevention and inspection

In order to ensure reliable crane operation and prolong the crane's life, each component on the crane must be operated correctly, and a proper maintenance schedule must be followed. To keep up the operating status and operational safety of the crane, please accomplish the maintenance of the crane at regular intervals listed in the rear of the chapter. The crane routine maintenance and inspection can be started with the following aspects.

2.7.1 Summarize

The inspection is used to judge whether the crane is in a safe and reliable state. It is basically a visual inspection and functional test. If an adequate assessment is not possible by this, further tests have to be taken, e.g. nondestructive testing. If necessary, disassemble the crane parts to observe the hid parts. The description of the crane components disassembly and re-assembly is listed in Chapter 3. When performing inspection and maintenance on the crane, the appropriate preventive actions should be taken according to TABLE2-14.

The abnormal temperature, sound, or libration can be the omen of failure, abrasion or mangle. The following sections list the detail.

Table 2-14： Preventive action (mechanical)

Inspection Items	Contents of Inspection	Preventive Action
Gear Gear coupling	Grease on gear Alignment Gear engagement and wear	Apply grease if poorly greased. Realign if out of tolerance Adjust the attachment if engagement is not proper. Replace excessive wear ones
Sliding surface of bearing, sheaves, wheel, pins, lever pins, etc.	Lubricating condition	Apply if necessary so that respective lubrication points are lubricated as required.
	Wear condition	Replace if wear is excessive
Bolts, nuts, keys, pins, shaft coupling bolts	Looseness, Attachment condition of split pin retainer	Tighten sufficiently. Replace if quality is deteriorated
Gear box	Lubrication Temperature sound	Add oil if the amount of oil is low. Replace if quality is deteriorated. Check abnormal sound
Wire ropes	Lubricating condition, broken wires, abrasion	Apply grease as necessary if grease is insufficient or degraded. See Section 4.3 for wire rope inspection.
Wire rope end termination	Condition of wire end	Tighten loose bolts with specified torque. See Section 4.2.3
High strength bolts, and welding for steel structure	Looseness Crack, damage	Replace if loose. Repair if crack or damage is observed.
Brake lining	Wear, fitness to brake disc	Replace if wear is excessive.
Brake disc	Wear of sliding surface	Repair if sliding surface is worn or rough.
Track and track clip	Bolt looseness, Track abrasion, Track distortion	Tighten loose bolts with specified torque, Replace excessive wear ones, Realign
Hook, headblock, lock pin in the spreader	Abrasion, crack	Replace excessive abrasion ones or ones with crack
Sheave, drum, idler	Abrasion	Replace if wear is excessive
Other consumable parts(such as seal, bulb）	Abrasion, damage	Replace if wear is excessive, Repair or replace damaged
Steel structure	Crack, paint, etc.	Refer to Steel structure maintenance manual
Safety device	Any abnormity	Repair while there is abnormity.

The crane and its additional equipment, primarily the safety devices, have to be checked for proper condition and function depending on work use. It requires special supervision of: − emergency-stop button and control; − drive & brake; − electrical equipment; − limit switch and overload protection.

2.7.2 Inspection of temperature rise

Temperature rise is a phenomenon. Maintenance personnel who well understand the temperature rise condition can judge the working condition of the crane. If there are no otherwise specifications, the temperature rise of the bearing should not exceed 60 ℃ (140℉) and the absolute temperature of the oil should be less than 85 ℃ (185℉). Infrared thermoscope is in common use. See Figure 2-30. The following temperature raise locations on the crane should be carefully inspected:



Figure 2-30 Infrared thermoscope

(1) High-speed rotation supporting locations, such as the motor bearings and the high-speed shaft bearing of reducers mainly used in main hoisting and trolley.

(2) Heavy-load and slow speed supporting locations, such as reducer low-speed shaft bearing, drum bearing pedestal, wheel bearing and sheaves bearing.

(3) Hydraulic actuator temperature rise, such as the hydraulic thruster of the brake (especially hydraulic thrusters of main hoist and trolley travel mechanisms), the relief valve and the throttle valve.

(4) Hydraulic oil reservoirs, such as the multifunction pump, emergency brake, gantry wheel brake, and sling hydraulic units.

Normally, the inspection of temperature rise condition can be achieved by hand touch and judging if the temperature is abnormal. When there is abnormal temperature rise, it is often accompanied with abnormal noise. If there is abnormal temperature, specialized thermometer should be used to measure and to aid in finding the cause of temperature rise. Only after the problem is solved, should the operation be resumed.

Generally, the causes of temperature rising are:

- ☞ Improper lubrication
- ☞ Improper gear meshing
- ☞ Bearing clearance is too loose or too tight (or bearing damage)
- ☞ The alignment coupling can't meet standard.
- ☞ Brake not fully released or not centered (one side released but the other side still touching the disc)
- ☞ Hydraulic thruster oil internal leakage
- ☞ Abnormal operation of the relief valve

2.7.3 Inspection of abnormal noise

The main cause of mechanical noise is vibration, impact noise, friction noise and structural transmitting. Maintenance personnel who understand the noise change can judge the operating condition of the crane through daily noise observation. Generally, the abnormal noise may be caused by improper gear meshing, bearing damage, loose bearing clearances, shaft bending, loose bolts, interference or motor fan damage, lack of lubrication, etc.

Noise can be identified/isolated by using an industrial stethoscope.

Noise measurement directly with ear may be harmful to ears, so only do this when necessary and for a short period.

Generally, the abnormal noise and temperature rise happen simultaneously, such as when the bearing clearance is too big or the brakes are not fully released, and so on. Therefore timely noise inspections can avoid many equipment failures.

2.7.4 Inspection of abnormal vibration

Vibration is usually related to noise and element fatigue. Close attention should be paid to the abnormal vibration during crane inspection, such as for the motor base, gear reducer base and drum support rigidity, base rigidity, bolt pre-tightening, misalignment of motor and gear reducer, dynamic balance of coupling and static balance of drum.

The inspection sequence is generally started with the check of alignment (radial and axial vibrations). If the vibration still exists after adjustment, the coupling should be removed and the motor checked separately.

After causes of the motor and loose bolts are eliminated and dynamic balance of coupling are checked, the coupling can be connected and the high-speed shaft

bearing of the gear reducer and should be driven and checked. Generally, the problems caused by improper gear meshing and lack of adequate rigidity of the installation base can be found during test running.

If the vibration happens after the crane is put into operation for a certain period, loss of alignment, loose bolts, excessive bearing clearance or improper gear meshing usually cause the vibration problem.

2.7.5 Inspection of crack

Crack inspection sometimes can prevent failure or disaster of a crane at its early phase. The inspection of crack consists of inspection of the driving parts, the connecting parts and the structural parts. The cracks are normally caused by fatigue and happen on locations of fatigue source, stress concentration, and repeated stress effect.

(1) Driving parts: such as driving shaft, wheel, gear, sprocket, sheave, coupling, braking wheel (disc), bearing and so on.

(2) Connecting parts: such as fixed shaft, wheel shaft, sheave shaft, structural connecting pin shaft, headblock connecting pin, spreader twistlock and so on.

(3) Structural parts: welding seam and base material.

The method of structural inspection can be visual inspection, which can be judged through the surface paint film. If the base material or welding seam is cracked, the paint film will be also cracked. Telescope can be used to inspect pre-determined regularly inspecting locations. If there is any suspicion after visual inspection, MT or UT inspection can be applied. 🕮Please refer to the Steel Structure Maintenance Program for details. Driving or connecting parts can also be checked during maintenance when they are knocked down.

2.7.6 Inspection of rail & rail clips

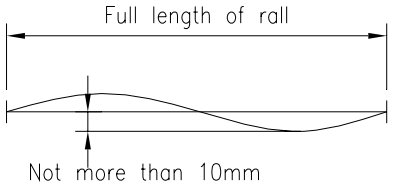
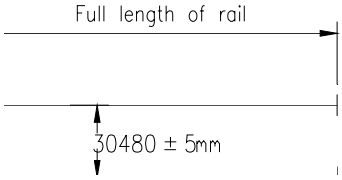
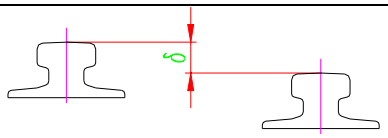
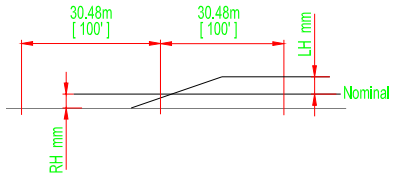
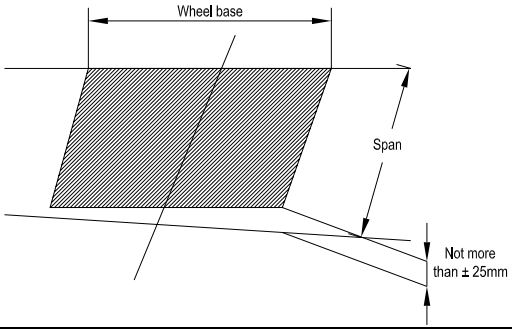
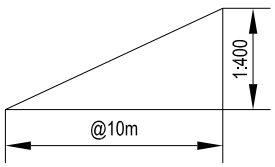
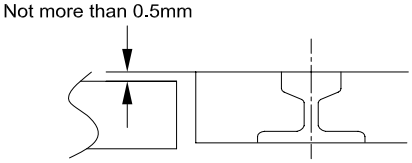
2.7.6.1. RAIL INSTALLATION TOLERANCES

When designing a complete container handling system, the problems of the crane rail track never appear to be insignificant, and that is important to crane's using.

Failures of rails, to a great extent, can be attributed to improper lying of rail during foundation work or the crane or trolley may not be tracking accurately. If the crane is used with curved or uneven leveled runway rails, its structural members can be twisted causing high stresses. Accordingly, the bolts in joints may be loosened and the bases of the various machines on the steel structure may be distorted. Thus, crane performance may be impaired.

In order to prevent such troubles, at the time of annual inspection, measure the level, straightness, span, etc. of the trolley travel rail and gantry travel rail and keep their records. Also, carefully inspect the trolley travel rail and gantry travel rail for cracks. If measured dimensions are not within the allowable tolerance in Table 2-15 or Table 2-16, immediately repair or replace the gantry or trolley travel rail.

Table 2-15：Allowable Tolerances for Gantry Rails

Inspection items	Measuring Method		Maintenance tolerance
Straightness		Measured by a transit or piano wire	≤±10mm [±0.393 in]
Span (S)		Measured by a steel tape measure	≤±5 mm [±0.197 in] (Check every 10m/32.81ft)
Height difference between rails		Measured by a transit	≤152±15mm [±0.59 in] (152mm is original dock design high difference)
Bend of rail left and right		Measured by a transit or piano wire LH+RH< Tolerance	≤20 mm [≤0.787 in] per 30.48 m [100 ft]
Levelness	In the nominal span and wheel base of the crane no point on any rail will be greater that ±25mm out of the common plane established by the other three corners of the crane. 		
Inclination		Measured by a transit	≤$\frac{1}{400}$ (Check every 10m/32.81ft)
Joint difference		Measured by a steel ruler	≤0.5 mm [≤0.02 in]

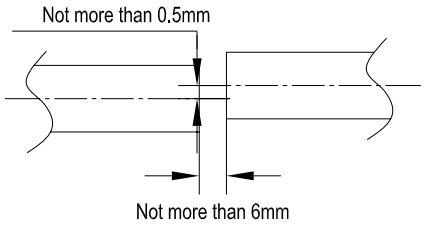
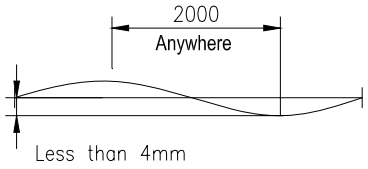
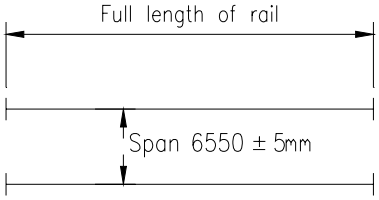
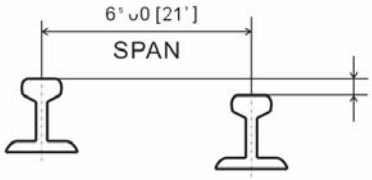
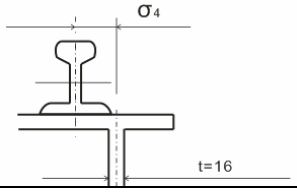
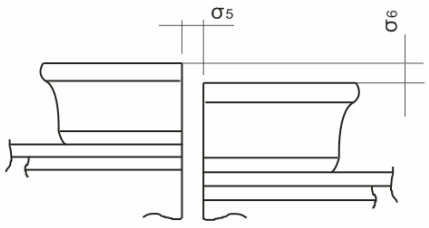
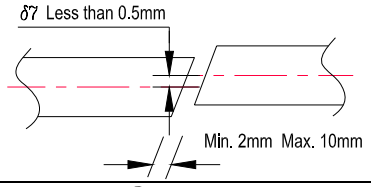
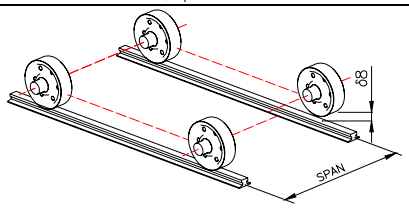
Inspection items	Measuring Method		Maintenance tolerance
Joint gap & deviation		Measure by a steel ruler	Gap≤6mm [≤0.24 in] No contact even in Summer. deviation≤0.5mm [0.02 in]

Table 2-16： Permissible Tolerances for Trolley Travel Rails

Inspection items	Measuring Method		Maintenance tolerance
Lateral Straightness		Measured by a transit or piano wire	≤4 [0.157 in]
Span (S)		Measured by a steel tape measure	≤±5 [±0.197 in]
Difference in height		Measure by a transit or a water level	0.15% of SPAN ≤9.6 mm [0.378 in]
Slope or grade	Less than1/200 {in the area without camber when there is not trolley. in the area with camber when there is trolley with full load. *Refer to Boom and Girder structural drawing for camber information.		
Center distance between rail and web		Measured by a steel ruler	σ_4≤6.0 [0.236 in]
Difference between boom and girder rail		Measured by a steel ruler	2≤σ_5≤10 [0.08 in] [0.39 in]
		Measured by a steel ruler	σ_6≤1 [0.0394 in]

Inspection items	Measuring Method		Maintenance tolerance
Center distance between boom and girder rail		Measured by a steel ruler	0≤σ7≤0.5 [0.08 in] [0.39 in]
Levelness		Measure by a water level	0.1% of span σ8≤6.4 [0.252 in]

2.7.6.2. COMMON FAILURES OF RAIL SUPPORT, RAIL CLIP AND RAIL INSTALLATIONS

1. Rail support system:

1) Concrete support: disintegration of the concrete (loosening of the holding down bolts due to grout or concrete failures).

2) Rail compaction devices:

a. Loosening of the bolts

b. Breakage of bolts, studs, or weld fillets

c. Elongation of holes

d. Rotation of clips

2. Rail:

1) Excessive wear or flattening of the rail head

2) Wear of edges of rail head

3) Rail breakage

4) Deformation at joints caused by wheels running over gaps

Causes of dilapidation

Figure 2-31 shows the acting force of the trolley. The crane operation causes repeated wheel loading of the rail resulting in fatiguing cycles. Fatigue failure occurs due to:

① High stress in the rail support,

② Rail movements causing fatigue in the rail attachment components.

Figure 2-32 and figure 2-33 shows the main abrasion form.

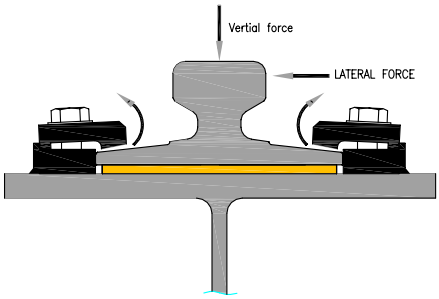


Figure 2-31：Rail under stress

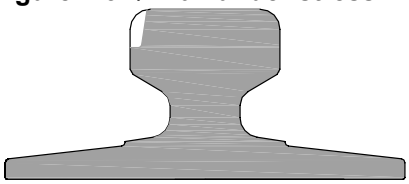


Figure 2-32：Side wear of rail

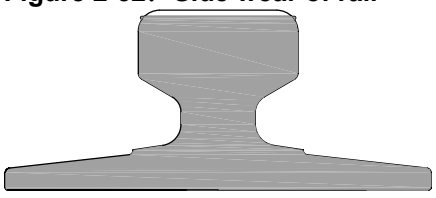


Figure 2-33：Surface wear of rail

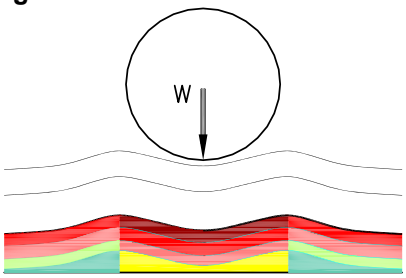


Figure 2-34：“bow wave” effect

High stresses caused by:

- Wheel flanges riding on the rail
- Compression stress due to concentrated wheel load
- Bending stress due to eccentric loading
- Lateral stress due to “crabbing”, trolley motion and swinging live load.

Rail movements caused by:
Longitudinal due to:
- Bending of the girder under load
- Thermal expansion
- Crane gantry or trolley accelerating and braking

Lateral due to:
- Crane or trolley "crabbing"
- Swinging of live load
- Gantry or trolley accelerating and braking forces

Vertical due to:
- Rail "bow wave" effect under wheel load
- Lifting of rail at interface of two adjacent girders
- Twisted rails

Rotation due to
- Twisted rails
- Load eccentricity

- Steering effects of crane on rail head

2.7.6.3. TROLLEY RAIL CLIP INSTALLATION AND INSPECTION

Inspect the bolts of rail clip to estimate whether they are loose. Screw down bolts if needed, and insure the exact position of stud. The process of installation about ZPMC trolley rail clip is listed hereinafter.

1) Welding of base. Weld the base all around with low hydrogen electrodes and 4mm fillet welding. Spot welding first, and then weld the base. Especially pay attention to clamp the base when welding to avoid the moving in upright direction and keep the contact of base. Refer to figure2-35.

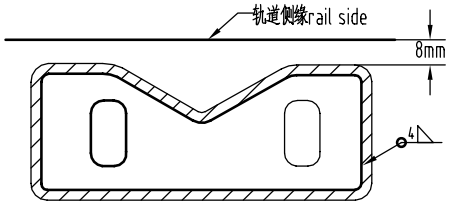


Figure 2-35：Welding of base

2) Installation of bolts and clip. Set the bolts into the holes of clip base, circumvolve it 90° and keep it lying in the wimble holes reliably. After confirming the bolts are oriented reliably (pay attention to the orientation line in the section of bolts), install clip and screw down nut manually. (It's important to screw down manually to ensure the correct location of the head of bolts in its head room.) Refer to Figure 2-36.

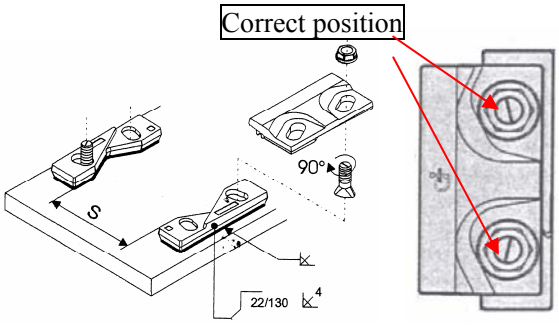


Figure 2-36：Installation of bolts and clip

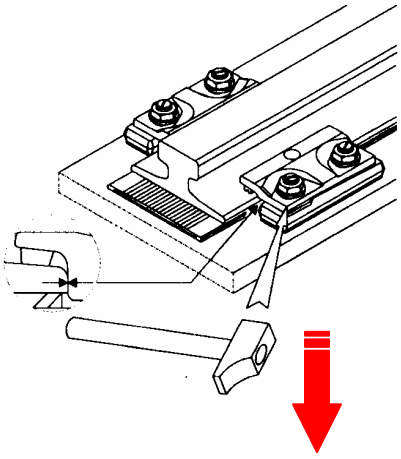
3) Cleaning up of base bolt hole. Clean up the rubbish (with air pump) and slag (with welding hammer) in the base bolt holes before set up the clips. Refer to figure 2-37.

4) Tighten the clip. Knock the clip slightly with hand hammer to keep the close contact of clip and rail side. Then, complete the tightening of the nuts to **150Nm**. In case of further painting protection, the use of solvent is prohibited. Refer to figure 2-38.

If, for any reason, a nut must be loosened after mounting, you must verify the correct position of the bolt shaft before re-tightening the nut.



Figure 2-37 Cleaning up



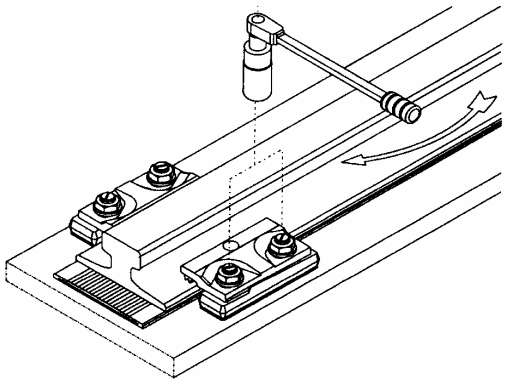


Figure 2-38： Tighten the clip

2.7.7 Inspection of gantry & trolley travel wheel

The gantry travel and trolley travel are indispensable for operation by this machine. Therefore, it is important to maintain the travel wheels and rail in normal state all times. The following passages describe the inspection procedure of the wheel.

The following items should be checked during the daily, monthly, and annual periodical inspections.

(1) Wear in treads and flanges

(2) No flanges riding on rail

(3) During traveling, both sides moving smoothly in parallel

(4) Abnormal noise or vibration

(5) Wear in wheels, bearings, etc., developing evenly for both legs.

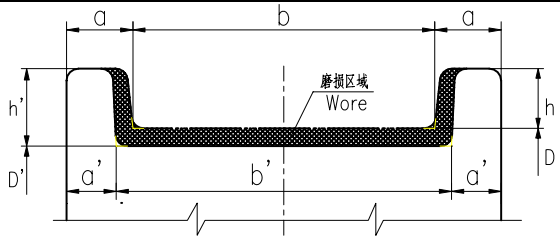
(6) Cracks or repair welding cracks in wheels.

(7) No break in rail flange. No condition of more than 3 places disfigurement with area over 3cm2 and depth over 30% of the thickness of the rail.

After inspection, if the wear of the gantry or trolley travel wheel reaches the values indicated in Table 2-17 or Table 2-18, replace the wheel.

The items listed above should be checked periodically in peacetime, monthly and yearly.

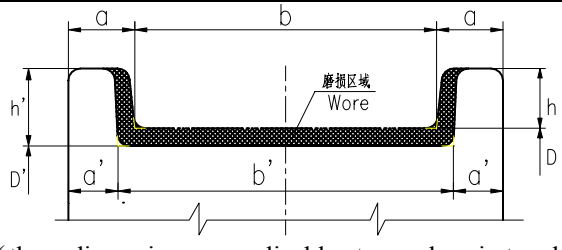
Table 2-17： Wear Limitation of Gantry Wheel



（these dimension are applicable at any place in tread）

Original dimension		**Wear Limitation**		
		Driving wheel		Driven wheel
D	800	D’	780 or the difference of the 2 driving wheel diameter is more than 2mm（0.2% D）	780 or the difference of the 2 driving wheel diameter is more than 5mm（0.5%D）
h	25	h’	35	
b	147	b’	165	
a	26.5	a’	17.5	

Table 2-18： Wear Limitation of trolley Wheel（draught trolley）



（these dimension are applicable at any place in tread）

Original dimension	**Wear Limitation**

D	630	D	615
h	25	h’	32.5
a	26	a’	18.5
b	100	b’	115

No wheel flange riding on rail is permitted. It will increase the travel resistance force up to 3 times as normal operation and will overload its driving system. If flange riding is observed, check the gantry drive coordination or the waterside trolley tow rope length agreement.

Severe wheel flange riding on rail will cause damage to the wheel and rail. Eliminate flange riding right away if it exist.

2.7.8 Inspection of rope sheaves, drums, idler and slap block

2.7.8.1. SHEAVE

If the following symptoms are found, check the wire rope for wear and replace the sheave at the same time of replacing the wire rope.

(1) The flange is damaged or deformed

(2) The wear in the flange is 10% or more of the diameter of the wire rope used.

(3) A trace of the wire rope lay can be observed at the bottom of the sheave grooves

(4) The wear in the sheave groove is 15% or more of the diameter of the wire rope used.Refer to figure 2-39.

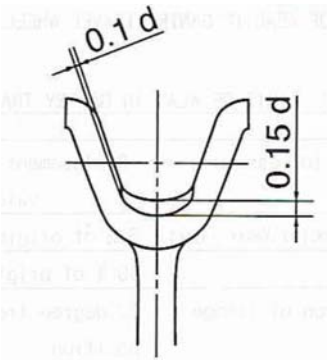


Figure 2-39 limits of wear in sheave groove

Refer to Figure 2-39 for limits of wear in sheave groove. The symbol “d” indicates the diameter of the wire rope used.

2.7.8.2. DRUM

Periodically check the drum groove for corrugation and excessive wear.

The wear limit in the drum groove is 20% of the diameter of the wire rope used. Refer figure2-40.

There should be no crack in drum groove, wire rope root point and drum axis.

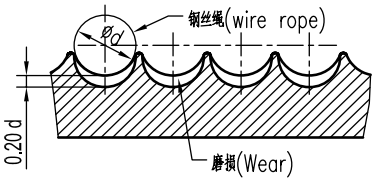


Figure 2-40 Limits of wear in drum groove

2.7.8.3. INSPECTION OF ROPE SUPPORT ROLLER AND SLAP BLOCK

The inspection of the rope support roller is described as follows:

(1) Verify that roller rotates freely.

(2) Determine the anti-friction material is peeling, aged or worn out. The wear resistant roller ring shall be replaced when the amount of wear reaches 60% of its original thickness. Make sure that the worn roller ring has been replaced before the wire rope penetrates the thickness of the roller to expose the base metal of the roller shaft. If this happens, the wire rope will rub against the metal to cause the abrasion failure of both the wire rope and the roller shaft.

2.7.9 Inspection of the spreader

As a direct performer of loading container, the load handling device’s reliability is very important. Please refer

to the ZPMC load handling device manual to get the information about the detail of the load handling device's maintenance standard. Following routine inspection to the load handling device are suggested in handing over or taking over:

1) The reliability of lock pin or shaft pin connected with upper-hanger and load handling device.

2) Check the structure in the headlock and spreader to eliminate damage, incomplete fusion or weld crack in the parts easily to be stroked. Check lubricating condition on the slide surface of flex arm and slide rail.

3) Lubricating and movement condition of key parts such as twist lock pin and peak-pin, reliability of twist lock pin locknut.

4) The tightness of all the limit-board of spreader, the integrity of limiter, the correctness of signal.

5) The reliability of cable used in spreader, and whether there is sudden damage.

6) The abrasion and damage condition of the head of twist lock of spreader.

7) The damage condition of guide-board of spreader.

8) The oil leakage condition of hydraulic system of spreader, and the oil surface location in the tank.

2.7.10 Inspection of hook and twist lock pin

The hook and twist lock pin become weaker against crack due to wear and hardening caused by extended use. Therefore, perform the following inspection once a year.

2.7.10.1. Hook

Perform the magnetic flaw or die check. After inspection, immediately make appropriate repairs or replacement:

1) Any bending or twisting by 10°.

2) Any distortion causing an increase in throat opening of 15%.

3) Any wear exceeding 10% of the original section dimension of the hook or its load pin.

4) Any self-locking hook that does not lock.

5) Any latch that does not close the hook's throat.

6) Repair of cracks, nicks and gouges shall be carried out by designated person by grinding longitudinally; following the contour of the hook provided no dimension is reduced by more than 10%.

7) Deform or crack on the screw segment of the tail of the hook.

2.7.10.2. Twist lock pin

Perform the magnetic flaw detection or die check. Repair or replace the parts on condition listed hereinafter.

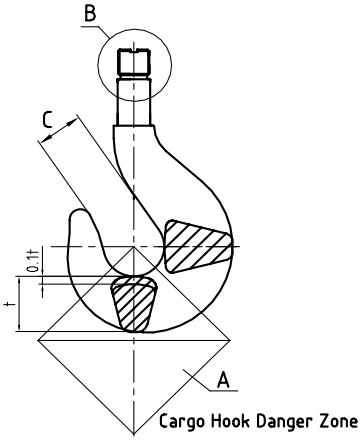


Figure 2-41 Annual check of cargo hook

1) Crack, crevasse or hair-crack on the face of the head of twist lock pin, shaft and the connect root of the lock head;

2) Damage or wear of the head of twist lock pin;

3) Deform or crack on the screw segment of the tail of the pin.

Always Inspect A and B area with special care!

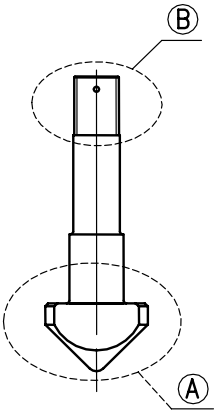


Figure 2-42 Annual check of Twist lock pin

It's not permitted to repair the disfigurement in the hook and twist lock pin by welding. Replace the failure part if there is un-repairable disfigurement.

2.7.11 Inspection of hydraulic handspike brake

Brake should be inspected periodically with items listed hereinafter:

1) Whether the brake lining and brake disk is clean and not covered with oil dirt.

2) Whether the arm bars of the brake are flexible.
3) Inspect whether the belleville spring is broken or hydraulic oil is leaked to the belleville spring through inspection window of the brake cylinder. Repair or replace the failure part if the break or leak happened.
4) Inspect whether the push rod of the limit switch is loose and the action is right.

Attention! Hydraulic handspike brake cylinder has especial structure inside and connecting proprietary technology. So don't disassembly it yourself and avoid belleville spring collapsing and hurting the people. Replace the fault cylinder as one piece.

5) Inspect the abrasion condition of the brake lining. If the attrited thickness of the brake lining is over 40% of the original thickness of the friction material, a new brake lining should be set.

2.7.12 Inspection of brake disc

DANGER! Any grease on the brake disc will cause the substantive drop of the brake torque.

Inspection of the brake disc is very important to the safety of the crane operation.

Periodically check if the brake disc is contaminated. Any oil or grease on the disc must be cleaned with detergent or solvent immediately.

Also check the brake disc surface roughness. If the disc is damaged, scoring or cracked, if anything is found as shown in Table 2-19, the brake disc should be replaced.

Table 2-19： Standards for Brake Disk Changing

	Main hoist	Trolley drive	Boom hoist	Boom hoist emergency brake	Main hoist emergency brake
Brake disc thickness	30mm	30mm	30mm	36mm	36mm
Wear limitation	27mm	27mm	27mm	32mm	32mm
Convex or concave on the disc surface	2mm	2mm	2mm	3mm	3mm
Brake lining wear limit	Attrited thickness should not more than 40% of base thickness. Interface between brake lining and brake disk should not less than 75%. The brake lining with adust face or aging should be replaced.				

2.7.13 Inspection of shaft and axle sleeve

Some parts of the crane, such as stay bar pin and handle pin of gantry anchoring is equipped with shaft + axle sleeve (copper sleeve commonly, sometimes copper sleeve with plumbago lubrication)。 Check if the pin and bushing are well lubricated and rotates relatively freely. The relative rotation always happens between the bushing and the pin. The bushing should stand still in the housing, so no wear should occur on the outer diameter of the bushing. The abrasion form of the shaft and axle sleeve and the measure should be taken are listed hereinafter.

1. Sort according to the abrasion form which is shown in figure 2-43.

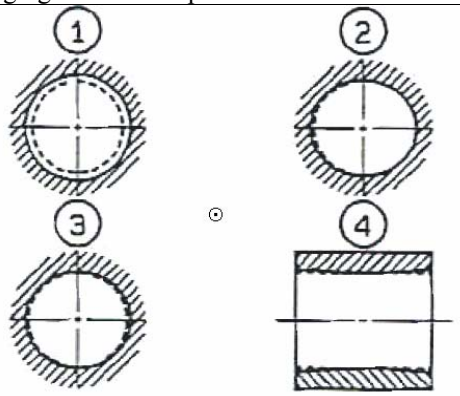


Figure 2-43 Abrasion form of shaft hole

1) symmetrical wear
2) wear induced by unilateralism load
3) wear induced by bidirectional load
4) wear only on the both sides

2. Measure while the hole is worn.

1) replace the worn axle sleeve
2) deal with the worn shaft holes conform to the steps listed hereinafter

a. bore the hole after repair welding
b. enlarge the diameter of pin, axle sleeve, and shaft hole
c. replace axle sleeve

If the wear of the bushing reaches the amount shown in Table 2-20, replace it.

Table 2-20： The Wear Limits of Bushings

D(mm)	Wear limitation of dia. (mm)	Location	Original dia.(mm)	Wear limitation of dia. (mm)
10~24	0.8	-	-	
25~40	1.2	-	-	
41~60	1.6	-	-	
61~100	2.0	-	φ90/φ70	~/φ72
		-	φ85/φ70	~/φ72
101~160	2.5	-	φ180/φ120	~/φ122.5
161~250		-	φ200/φ180	~/φ183
		stay bar	φ260/φ230	~/φ233

2.7.14 Inspection of oil seal

Please check the oil seal if leakage is founded according as the description listed hereinafter. .

1) whether there has been appropriate seal set in the leakage point
2) Whether the brim seal is damaged by keyway, spring, screw, rough surface or the brim of shaft
3) Whether the seal is set in reverse direction.
4) Whether the deformation of seal is caused by setting with unsuitable tool.
5) Whether the venthole is jammed.

2.7.15 Bearing unit inspection

In the crane's working, judge the state of the bearing housing during operation, then take the following measures if abnormality is observed.

(1) If vibration is found:

If the vibration is severe, first check the retaining bolts for looseness and verify that there is no looseness between the bearing unit and the installation base. Tighten the bolts as necessary. If the vibration remains after the above mentioned procedure, check if the causes of other rolling element vibration such as bent shaft exist. If no abnormal part is found, replace the bearing unit since the excessive clearance due to wear inside the bearing is suspected.

(2) If abnormal temperature rise is found.

If the temperature of the housing rises over 20°C (36°F) after subtracting the ambient temperature, carry out the inspections below.

1) Check if any causes of the temperature rise exist, such as no relief clearance due to the one-side free bearing, overloading, etc.
2) If no abnormal condition is found, replace the bearing since the severe condition such as an initial state of bearing damage etc. is suspected. Especially the temperature rise with abnormal noise is considered to be a sign of developing damage. Replace the bearing immediately in such a case.

(3) If abnormal noise is found.

If abnormal noise is heard from the bearing unit, carry out the inspections below.

1) Measure the bearing block temperature. If the temperature is less than 20°C (36°F) after subtracting the ambient temperature, check the retaining bolts for looseness and verify that there is no looseness between the bearing unit and the installation base. Tighten the bolts if necessary.
2) If the abnormal noise remains after procedure 1, the following causes are suspected. Immediately replace the bearing judging from the abnormal noise degree and the machine importance.
   - ◆ Slight wear appears on the track surface and/or rolling element of the bearing.
   - ◆ Inclination toward noise.
   - ◆ Foreign materials inside the bearing
3) Measure the bearing block temperature. If the temperature is over 20℃ (36℉) after subtracting the ambient temperature, replace the bearing unit since the developing peeling of the track surface or rolling element is suspected.

2.7.16 Inspection procedure of key and keyway

The key and keyway suffer various kinds of damage according to the used condition. If and damage is found on inspection, immediately replace the key. The damage types

and their cause are as follows:

(1) Shearing of key

The key may be sheared if the abnormal torque that is larger than key strength is applied in such a case. The keyway often may be deformed as well, so re-cut the keyway and install the key with hardness of superior grade.

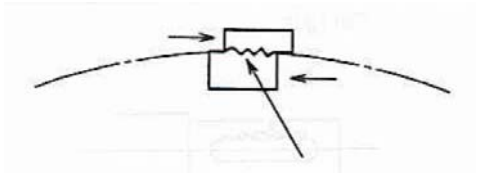


Figure 2-44 Shearing of Key

(2) Crack of keyway

The stress concentrates on the edges of the shaft or the keyway in the fillet engaged with the shaft. Therefore, cracks may be caused if the part suffers abnormal torque larger than their strength.

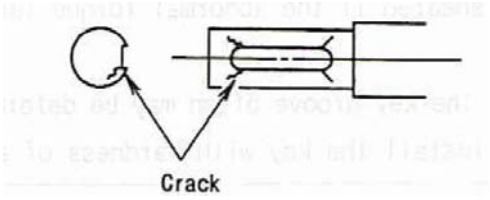


Figure 2-45 Crack of Key Groove

If the condition remains, the crack may be enlarged, resulting in keyway damage of shaft failure.

(3) Wear of keyway side

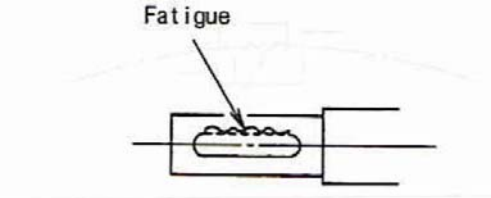


Figure 2-46 Wear of Key Groove Side

The fatigue and deformation may be caused if abnormal torque is applied to the keyway sides and abnormally high surface pressure is struck on the keyway sides repeatedly.

2.7.17 Inspection of consumable parts

Consumable-parts are those parts which may be depleted or worn out by use of the crane and **they do not belong to guaranty products**. During crane maintenance and inspection, pay special attention to the consumable parts listed as follows:

1) Inspection of wire rope for abrasion, wire breaks and other damage. ☞ Refer to the relative provisions in Chapter 4.
2) Check if there is any oil leakage from gear seal rings.
3) Inspection of braking shoe abrasion.
4) Inspection of carbon brush of conducting ring (such as DC motor and spreader cable reel slip rings).
5) Inspection of kryptol for trolley rail and wheel lubrication.
6) Inspection of all lamp bulbs.
7) Make sure the spare fuses are available.
8) Inspection of filters.
9) Inspection of vulnerable parts, such as the spreader flippers.

2.7.18 Inspection of steel structural members

	**The structural integrity of the crane requires that no attachments or fixtures of any kind be welded to any part of the crane without prior written approval of the responsible Structural Engineer.**
	**Please refer to Steel Structure Maintenance Program for more inspection information.**

In order to maintain the crane in good condition, continuing inspection is very important. When damages occur to steel structural members, this crane may be unusable and require detailed inspection. Not only machine damage but also accidents including human injury may arise.

Deformation of the steel structure members may be caused by improper usage of the crane, development of cracks, and loose bolts. Therefore, always follow the operating procedures described in the container crane operation instruction manual and conduct the regular inspections as described in this manual.

Inspection and diagnosis methods for metal structure:

1) Inspect and judge the metal structure by human sense and experience:

- Check to find out whether the hinge points and

the interconnecting pieces run smoothly and flexibly, whether have abnormal sound and blocking phenomenon, whether have a good lubrication and a severe abrasion, whether the center axle have movement and so on.

- Check the bolts connection to eliminate loosening.
- Check the main connection pieces and welding to find out painting crimple, spalling or welding cracking.
- Check the components to find out global or local deformation and bending, folding twisting and damage phenomenon.

2) Period inspection through non-destructive detection should be carried out for the welding of FCM pieces (for detailed information refers to STRUCTURE AND MAINTENANCE MANUAL for QUAYSIDE CONTAINER CRANES).

3) Use electrical logging stress method to test the strength and rigidity of the metal structure. Through careful and quantitative inspection to make a scientific judgment on the working condition of the metal structure.

The following things should be paid much attention to:

1) Check deformation, corrosion, crack, loose bolts, etc. at the time of regular inspection such as daily, monthly, and annual inspection.

2) The deformation of steel structure can degrade the alignment of the machinery installed on the structures, increasing operating noise, causing improper engagement of the gears, etc.

3) If some deformation of steel structures, development of cracks, loose bolts, etc. is observed, immediately report to the service office, and repair it under the supervision of the structural engineers.

4) In order to prevent corrosion of the steel structure, clean clogged drain holes. Repaint areas if there is any indication of rusting. Corrosion will reduce the thickness of the structural members, increase members stress, and may create cracks. In order to inspect cracks in the structural members, thoroughly remove the rust and paints from the area, then conduct the indicated testing.

5) The lubrication of the stay bar pins is essential to keeping the stay bars sound. If the rusting of the pin occurs on the stay bar friction surface, the additional bending moment will be induced on the bar and may cause the initiation and propagation of cracks. Therefore, the lubrication must be made periodically to remove old grease and to refill the new grease.

2.7.19 Inspection of all safety protections

Safety protections should be checked before any work.

1) Regularly inspect if the braking torque setting-up of all brakes. Make a practical plan to periodically verify the braking torque. All slow-down limit switches and over-travel limit switches. Ensure that the positions of all encoders, functions of all interlock protections are correct.

2) Safety inspection of elevator according to its manual。

3) Regularly inspect the hoist over-travel limit switches and hook of service crane.

4) Visually inspect if all buffers mounting condition and wheel stoppers are reliable.

5) Replace any missing or illegible warning sign.

6) Refer to Chapter No.1 to find out the working condition of crane safety equipments like the boom stowage device, gantry anchor device, tie down device and so on.

7) Check the function of the operator's cab, and check the headblock to find out crack and loosing parts. Check the ladder rails to find out the loosed or deformation bolts.



Never tamper with any safety protection devices. It's about your own safety!

2.8 Inspection task list

The inspection and maintenance tasks for cranes are listed in this book, which can be used as a reference for the owners. It should be declared that owners have more experence in inspection and maintenance than crane suppliers. We will be

honored to receive ideas and advices to improve the inspection and maintenance task list and make it more operational.

In this chapter, the inspection and maintenance tasks for cranes in regular service are divided into ten (10) general classifications based upon the intervals at which inspection/maintenance should be performed. The intervals in turn are dependent upon the nature of the critical components of the crane and the degree of their

exposure to wear, deterioration, or malfunction. The ten (10) general classifications are defined with respective intervals between inspection/maintenance as follows

(Table 2-21)：Period classification

No..	Classifications	Definition
1	PREOPERATION	Prior to the beginning of any crane operation.or after maintenance of the crane
2	SHIFT	Before, during and after work which equals to daily
3	WEEKLY	To be performed once per each calendar week or each 100 hours（the first arrival is accounted）。
4	BIWEEKLY	To be performed once per each calendar two-week period or each 150 hours（the first arrival is accounted）。
5	MONTHLY	To be performed once per each calendar month or each 300 hours（the first arrival is accounted）。
6	QUARTERLY	To be performed once per each calendar quarter of a year(three months period) or each 1,000 hours（the first arrival is accounted）。
7	SEMIANNUALLY	To be performed once per each calendar half year(six months period) or each 2,000 hours（the first arrival is accounted）。
8	ANNUALLY	To be performed once per each calendar year. (twelve months period) or each 4,000 hours（the first arrival is accounted）。
9	QUINQUENNIALLY	To be performed once per each five-year period(sixty months period) or each 20,000 hours（the first arrival is accounted）。
10	DECENNIALLY	To be performed once per each ten-year period(120 months period) or each 40,000 hours（the first arrival is accounted）。

(Table 2-22)：Preoperation period inspection task list (for queyside container crane)

Equipment No.：		Start working time：	Table No.：	
Checker：		Finishing time：	Running hours：	
Maintenance personnel：		Inspection date：	Confirmed by technician：	
No.	**Inspection items**	**Description**	**Result**	**Solvent**
Whole crane				
1	Cleanliness	Clean up garbage and tools left on the crane to avoid altitude drop.		
2	Marker	Hang "Preoperation period“marker.		
3	Emergency brakes	Avoid emergency brakes and other abnormal brakes (like electricity cut off) in preoperation period.		
4	Tightening	Inspect and checkout every connecting parts. Tighten or adjust if necessary.		
Machinery house				
1	Motors for the three main mechanisms.	Reduce 10%-15% of the rated load, low speed running for at least 50 hours, smoothly accelerate and decelerate. No emergency brake.		
2	Emergency brake (low speed brake)	Apply the brake umpty times under empty load or 50% load, watch the stabilization. **Focus of inspection** in preoperation period: 1.Joint condition of bush and brake disc; 2. Brake disc temperature should below 350 ℃;3. Loosing of screw connector; 4. Limit switch operation; 5.Running and brake sounds should be normal;6. No leak from the hydraulic system.		
3	High speed brake	Apply the brake umpty times under empty load or 50% load, watch the stabilization. **Focus of inspection** in preoperation period: 1.Joint condition of bush and brake disc; 2. Brake disc temperature should below 350 ℃;3.Loosing of screw connector; 4. Switch signal; 5.Running and brake sounds should be normal		
4	Hydraulic system	No leak from the pipes and valve manifold, reading from the gauge should normal, clean the oil reservior, filter and other valves each time the fluid is changed.		
5	High speed coupling for the three main mechanisms	No abnormal noises ,no loose bolt		
6	Low speed coupling for the three main mechanisms	No abnormal noises,no oil leak		
7	Gear boxes for the three main mechanisms	No abnormal noises or temperature rise, clean the box and filters each time the fluid is changed		
8	Electric room	Make sure no logic short-circuit or constraint. Check the readings from the guage on the electric cabinet		
9	Operating inspection	Run umpty times under empty load or light load to make sure running and braking are smooth		
Girder				

1	Multi-function hydraulic system	No leak from the pipes and valve manifold, reading from the gauge should be normal, clean the oil reservior, filter and other valves each time the fluid is changed.		
Gantry				
1	Gantry hydraulic system	No leak from the pipes and valve manifold, reading from the gauge should be normal, clean the oil reservior, filter and other valves each time the fluid is changedl.		
2	Rail clamps	Apply the brake umpty times under low speed; check whether the brake bush and rails are clean.		
3	Reducers	No abnormal noises or temperature rise, clean the box and filters each time the fluid is changed		
4	Gantry motor	Reduce 10%-15% of the rated load, low speed running for at least 50 hours, smoothly accelerate and decelerate. No emergency brakes.		
The "result" column should be filled in :[√]pass；[O] maintain/replace；[×]rejected. This sheet should be returned after inspection finished.				

(Table 2-23)：Shift task list

Equipment No.：		Start working time：	Table No.：	
Checker：：		Finishing time：	Running hours：	
Maintenance personnel：		Date	Confirmed by technician：	
NO.	**Inspection items**	Description	Result	Solvent
Whole crane				
1	Communication system	Works well		
2	Lighting	Check the walkway lighting, spot light and aviation alarm light. Make sure the aviation alarm light works well.		
3	Elevator	Make sure the elevator runs normally and smoothly, and stops at the correct floor.		
4	Air conditioner (machinery house and operator’ s cab)	Works well		
5	Emergency brakes	Works well at each emergency stop		
Machinery house				
1	Wire rope of the three main mechanisms (visual inspection)	Well lubricated, no loose press plate, reliable colligation for the redundant rope.		
2	Brakes	Keep reliable function, no looseness or deformation; Brake disc: proper clearance, no abnoamal wear or cracks, neat surface; Friction washer of the low-speed brake wears moderately, no abnormal wear on cylinder of the brake and disc spring, check the motor brake thrust stroke reserve; Friction washer of the high-speed brake wears moderately, Brake moment is proper, stroke reserve within green area.		
3	Hydraulic system	Proper oil level, No leak at pipe and valve manifold. Pressure in the specified range with pressures stable.No internal and external leak at low-speed brake.No leak at pipe fitting. Check oil temperature and abnormal noise during the running of the hydraulic pump.		
4	Drums of the three main mechanisms, high-speed coupling and low-speed coupling	No abnormity (visual inspection)		
5	Bearing Pedestal	No abnormity (visual inspection), no loose bolts, no abnormity at limit device at terminal		
6	Gear boxs of the three main mechanisms	Proper oil level, no abnormal noises, temperature rise or leak.		

7	Emergency brakes of electric room and machinery house	Work well.		
8	Electric room	Make sure no logic short-circuit or constraint. Check the noise and smell of the control panel; check the CMS log. verify the current and voltage from CMS and PQM		
9	Noise and vibration	Check the noises and vibration from motors, gear boxes, brakes, drums, couplings and bearings of the three main mechanisms		
10	Motors of the three main mechanisms	Motors and fans work well		
11	Doors and windows of the machinery house	No damage. Closed after work.		
girder				
1	Power cable chain system	No intertwist for the cables, no abnormal noise when the cable chain system runs.		
2	Trolley	No abnormal wear for all the Guide Idles on the trolley, rotate freely. Keep the wheels and horizontal rollers work well without noise.		
3	Wire rope	The wire ropes for main hoist and trolley are at correct location.		
4	Boom buffers	No aging, looseness;no deformation and cracks for the limit plate		
5	Slap block for stay support of boom	No deformation (visual inspection), no loose or missing blots.		
6	Multi-function hydraulic system	Proper oil level, No leak at pipe and valve manifold. Pressure in the specified range with pressures stable.No noises and vibration.		
7	Tensioning cylinder of trolley and catenary trolley	No leak at pipe connector, check the limit switches in right position. No internal leak for cylinders.		
8	The sheaves	No deformation, looseness and rotate freely for the sheaves on the boom tips and trolley girder, sheaves on the trolley and sheaves for the boom.		
9	The Guide Idle at the hinge point(visual inspection)	No abnormal wear		
10	Trolley rail(visual inspection)	No looseness, wear or separateness for clips and bolts on trolley rail. No impact when the trolley drive through the rail joint. No crack, incomplete fusion weld, looseness for the connecting plate and locating block on the short rail.		
11	Clamp plate for hoist wire	No looseness or deformation		

	rope at the boom tip			
12	Anti-collision limit divice	Check the anti-bump wire rope at the two sides of boom		
13	The sheavs on tension device and anti-snag device	In position.		
14	Spreader cable reel	The cables are in the groove, Check the noise and vibration of the motor and the reducer.		
“A”-frame platform				
1	Limit plate	No deformation or cracks. Work well with limit device		Check while booming up
2	Limit device	Well wired and no loose bolts		
3	Boom hoist sheaves and bearings	No abnormal noise, wear, deformation or looseness.		
Operator’s cab				
1	Control table	Check trolley: speeds, accelerations, AI speed reduce, stops, cable festoon system and boom motion. Verify main hoist: speeds, accerelations, upper/lower slowdown and stops; check the low speed hoist in ship clapboard; Check the lighting function in operator’s cab; Check the spreader guide plate and pin; Check the function of joystick and control panel.		
2	Air conditioner & electric system	Check the function of air conditioner, air filter, lighting, defrost device and windshield wiper		
3	Chair	Check the function of operator’s seat belt and shoulder harness.		
4	Doors and windows	Keep clean. Closed after work.		
Gantry cable reel				
1	Gantry cable reel	No missing, loose, twisted bolts on the reel; no abnormal wear for cable and reliable motion for the reel.(visual inspection)		
2	Reducer	Proper oil level, no leak or vibration		
Gantry				
1	Gantry hydraulic system	Proper oil level, No leak at pipe and valve manifold. Pressure in the specified range with pressures stable.Check the oil temperature and abnormal noises of the hydraulic pump.		
2	Cylinder of rail clamps and hydraulic pipes	No leakage, pipes are well fixed, brake works well		
3	Buffer	No deformation, leakage or damage		
4	Rail sweeper	Correct position and no deformation		
5	Cable roller and guide device	No looseness or abnormal wear; no block for the two springs of guide device		

6	Reducers	Proper oil level, no leakage		
7	Rail	No obstacle on the rail, no damage for the rail		
8	High voltage cable groove	No obstacle, no damage		
9	Motor	Check the velocity and acceleration		
10	Emergency brakes	Work well		
11	Stowage device	Apply the stowage after work		
Spreader and headblock				
1	Spreader	Check the function of the spreaders, flipper and twistlocks; Check the safety and loosing condition of driving chains;fix the spreader at the position of 20ft,40ft and 45ft; Check the oil level in telescopic driving reducer; Check the connection of the cable plug.		
2	Headblock	Check the assembly and function of the hoist rope slack limit switches.		
The "result" column should be filled in :[√]pass; [0] maintain/replace; [×]rejected. This sheet should be returned after inspection finished.				

(Table 2-24)：Scheduled maintenance task list (weekly)

Equipment No.：		Start working time：	Table No.：	
Checker：		Finishing time： ：	Running hours：	
Maintenance personnel：		Date：	Confirmed by technician：	
NO.	**Inspection items**	Description	Result	Solvent
Whole crane				
1	Lighting	Check the function of all working spot lights		
2	Limit device	Check the wire connection and fastenings		
Machinery house				
1	Wire ropes of the three main mechanisms	Well lubricated, no loosened press plate, reliable colligation for the redundant rope.		
2	Brakes	Keep reliable function, no looseness or deformation; Brake disc: proper clearance, no abnoamal wear and cracks, neat surface; Friction washer of the low-speed brake wears moderately, no abnormal wear on cylinder of the brake and disc spring, check the motor brake thrust stroke reserve.; Friction washer of the high-speed brake wears moderately, Brake moment is proper, stroke reserve within green area.		
3	Hydraulic system	Proper oil level, No leakage at pipe and valve manifold. Pressure in the specified range with pressures stable. No internal and external leakage at low-speed brake. No leakage at pipe fitting. The filter reads well.		
4	Drums of the three main mechanisms	No cracks, abnormal wear or loosened bearing pedestal; no noise, vibration and looseness for bearings; no incomplete fusion weld for lock block; check fastness of the press plate at rope tips.		
5	Bearing pedestals of the three main mechanisms	Check noises and temperature of the bearing; Tighten the bolts on bearing pedestals.		
6	Gear boxs of the three main mechanisms	Fastened well; proper oil level; no leakage.		
7	Low speed couplings of the three main mechanisms	No loosened nuts and keys; wear is within specified range.		
8	High speed couplings of the three main mechanisms	Elastic components work well; no loosened nuts and keys		
9	Limit device on the tip of the bearing pedestal	No looseness or abnormal abrasion at encode coupling		
10	Service crane	Run well; well lubricated; no loosened press plate; no abnormal for the wire		

		ropes		
11	Electric cabinet	Check the aeration condition and the fans.		
Girder				
1	Anti-snag platform	All slap blocks fixed well; no deformation, looseness or abnormal wear for fasteners of sheaves.		
2	Anti-snag hydraulic system	Proper oil level; no leakage at the pipes and valve manifolds; no internal and external leakage at cylinder; the filter reads well.		
3	Pin shafts of anti-snag	Well lubricated and no abnormal wear		
4	Hydraulic system of tension device	Proper oil level, No leakage at pipe and valve manifold. Pressure in the specified range with pressures stable.		
5	Fixing bolts for sheaves of tension device	No looseness or damage		
6	Tension cylinders of trolley	No leakage at pipe joints, check the limit devices in right position. No internal leakage for cylinders.		
Cable chain system				
1	Traveling chains	No departure, no deformation, normal abrasion;		
2	Fixing bolts and screws	No looseness; no broken screw.		
3	Dead block and rubber press block	No looseness,aging, deformation or crack		
4	Towing arm	No damage and departure.		
5	Cable clip	No absence, looseness, deformation or cataclasm; no broken cable		
Operator’ s cab and trolley frame				
1	Trolley front cable chain towing device	No deformation, looseness		
2	Operator’ s cab	Chair: clean, slide freely and reliable fixed; Door: open freely, the clearance to the locking plate is normal and no block for the locking plate.		
3	Sheaves	No deformation, looseness or abnormal wear		
4	Press plate for wire ropes	No deformation or looseness		
5	Guide Idles on the trolley	No abnormal abrasion,looseness; rotate freely		
6	Reducer for spreader cable reel	Well fastened , enough oil; no noise and glowing; Check the chain		
7	Each parts for spreader cable reel	Well fastened; normal distance between spreader cable reels, no abnormal abrasion for cable		
8	Trolley wheels and horizontal rollers	No abnormal abrasion; no scrape between the horizontal rollers and the bolts on		

		rail; no loosened bolts for covers; well lubricated and no abnormal condition for wheel bearing.		
9	Trolley buffers	No deformation, leakage or damage		
10	Linkage with trolley	Well lubricated wire rope with no broken strand, the diameter reduction is less than 7% due to abrasion; distance between rope lays where broken wires are found is less than 10 wires. No looseness, abrasion or departure for trolley rail clips and bolts. No obvious concaveness at the joint of trolley rails, just height and horizontal deviation. No obvious concaveness cracks or abnormal abrasion for trolley rails.No cracks, incomplete fusion weld or looseness for p in block and conneting paltes of short trolley rail. No loosened slap blocks under the machinery house and booms. No abnormal abrasion for the guide idle at the boom hinge, and the guide idle rotate freely.		
11	Plunger limit for main hoist	Exact contact, reliable connection for limit switch. No loosened bolts		
Along the boom				
1	Sheave block and bearings on boom	No abnormal abrasion,looseness; reliable function of sheaves		
2	Anti-collision wire rope	Normal tension or no damage		
3	The boom hinge	No abnormal noise or abrasion. Well lubricated bearings and fastened bolts		
4	Guide roller at the boom hinge	No abnormal abrasion, looseness; rotate freely; well fixed slap blocks.		
5	Boom buffer	No looseness, aging; no deformation or crack for the limit plate.		
6	Beam stays	Fasten the bolts and lubricate the pin shaft		
7	Slap blocks for beam stay support	No distortion; no loosened or missing bolts.		
8	Slap blocks for wire rope	Well fixed and no abnormal abrasion		
9	Fixed guide rollers	No abnormal abrasion; rotate freely, well fixed slap blocks.		
10	Trolley buffers	No deformation or damage		
11	Sheaves	No deformation for the fiexed sheaves at tips of boom and girder and the sheaves on the trolley frame; no abnormal for sheave shafts;well lubricated bearings		
Boom tip				
1	Clamp plate for hoist wire rope	No looseness or deformation		
2	Slap blocks and bolts	Well fastened		
3	Trolley and catenary	No looseness, deformation or abnormal		

	trolley sheaves	abrasion		
"A" -frame platform				
1	Hook limit plate	No deformation or cracks; well contact with limit device		Inspect after the boom raised
2	Rope slack limit shaft	Well lubricated, flexible and reliable		
3	Sheave block and bearing for boom hoist	No abnormal noise, abrasion, deformation or looseness		
4	Clamp plate and support for rope broken protection	No looseness, deformation or cracks		
5	Stay connecting plate and bolts for block plate	No deformation, incomplete fusion weld, looseness or abrasion		
Gantry cable reel				
1	Gantry cable reel	No missing, loosened and twisted bolts on the reel; no abnormal abrasion for cable; and reliable motion for the reel.		
2	Reducer	Proper oil level, no leakage		
Gantry				
1	Gantry hydraulic system	Proper oil level, No leakage at pipe and valve manifold. Pressure in the specified range with pressures stable.The filter reads well.		
2	Cylinder for rail brake and hydraulic pipes	No leakage and well fixed pipes		
3	Couplings	No loosened bolts or abnormal		
4	Buffers	No deformation, leakage or damage		
5	Stowage device	No deformation or incomplete fusion weld, well lubricated, flexible and reliable		
6	Rail sweeper	Correct position and nodistortion		
7	Cable roller and guide device	No looseness or abnormal abrasion; no block for the two springs of guide device		
8	Reducer	Proper oil level, no leakage		
Spreaders and headblocks				
1	Hydraulic system	Normal hydraulic pressure		
2	Hydraulic oil box	No leakage; reliable fixed and enough oil;		
3	Bolts	Check the bolts at shake protection device, flipper reducer pedestal and flipper motor and so on.		
4	Lubrication	Check the lubrication for flipper drive reducer, telescopic chain, bearings and twistlocks		
5	Cleanliness	Clean and fasten the sensors and limit devices		
6	Connection between spreader and headblock	Ensure the function of the limit switches for separating or connecting		

		the spreadr and headblock		
7	Junction box and control table on the headblock	Check the connection and fastening of the bolts		
8	Over-height frames	Well lubricated twistlocks and other active parts.		
The ″result″ column should be filled in :[√]pass；[0] maintain/replace；[×]rejected. This sheet should be returned after inspection finished.				

(Table 2-25)：Scheduled maintenance task list (bi-weekly)

Equipment No.：		Start working time：	Table No.：	
Checker：：		Finishing time：	Running hours：	
Maintenance personnel：		Inspection date：	Confirmed by technician：	
No.	**Inspection Items**	**Description**	**Result**	**Solvent**
Crane mechanism				
1	Brakes for the three main mechanisms in machinery house	Clean the brake disc; Check the abrasion of bush for the brakes		
2	Anti-collision device for boom	Check the position of anti-collision wire ropes and anti-collision limit device		
3	Steel structures	No abnormal damage		
4	Gantry rail brakes	Clean the brake pad; Check the abrasion of pad for the brakes		
Spreader mechanism				
1	Motors and couplings	Run well; no vibration or abnormal noise		
2	Gear box	Run well; no abnormal noise		
3	Hydraulic motors for telescoping and flipper	Reliable function; no abnormal noise, leakage or looseness; proper oil level.		
4	Hydraulic pumps and control valves	Reliable function; no abnormal noise or leakage.		
5	Hydraulic pipes and joints	No abnormal noise,leakage or distortion		
6	Cylinders for telescoping, twistlock, middle twistlock and hook	Reliable function; no abnormal noise,leakage or looseness; well lubricated		
7	Hydraulic oil box	Well fixed; no leakage; enough oil		
8	Hydraulic filter	No leakage and reads well		
9	Twistlock device	Reliable function; no distortion or damage;no crack for bush; no deformation for the winch;No loosened nuts		
10	Container landing pin	Reliable function; no distortion or damage; no excessive wear		
11	Flipper	Reliable function; no deformation, crack or abnormal damage; Bolts are good; No loosened nuts.		
12	Drive chains	Well fastened and lubricated, no distortion and abnormal wear; looseness for driving chain is 25mm～38mm(for the 4800mm-long chain)		
13	Middle twistlock slip slot(up and down,left and right)	Well lubricated; no abnormal abrasion		
14	Telescope beam	Well lubricated; no abnormal abrasion		
15	Main structure and structure of each part	No distortion, incomplete fusion weld, crack; no crack around the		

			twistlock hole		
16	Tip bottom roller		Well lubricated and flexible function		
17	Slap block for flipper and telescope beam		No abnormal abrasion or looseness		
18	Bolts fastening		Check the bolts on reducer pedestal and hydraulic motor of flipper.		
19	The sheaves, bearings and fasteners for headblock		No abnormal noise,abrasion, deformation or looseness		
20	Steel structure for headblock		No deformation,incomplete fusion weld, crack;		
21	Twistlock device for headblock		Reliable function, no deformation or damage; no crack for bush. no deformation for the winch;No loosened nuts		
22	Portable bar and pin shaft		Flexible function; well lubricated; no distortion		
23	Fixing device for spreader cable reel on headblock		Flexible function; no deformation or looseness; no deformation for spring and no crack for bearings		
24	Door for headblock		Open freely and block plate works well		
25	Sheave cover on headblock		No distortion, incomplete fusion weld or cracks		
electric device for spreader					
1	Junction box at two terminals		Well connected		
2	Power box and fast connecting plug		Well fixed and no damage		
3	Junction box on headblock		Well connected, no damage or water inleakage		
4	Limit devices	Middle twistlock limit	Act normally, well connected.		
		Middle twistlock position limit	Act normally, well connected.		
		Open lock limit	Act normally, well connected.		
		Close lock limit	Act normally, well connected.		
		Container landing limit	Act normally, well connected.		
		20FT limit	Act normally, well connected.		
		40FT limit	Act normally, well connected.		
		45FT limit	Act normally, well connected.		
		Twin-container inspection limit	Act normally, well connected.		

		Spreader and headblock connection limit	Act normally, well connected and sealed		
		Middle twistlock hook permitting limit	Act normally, well connected.		
5	Electromagnetic valve plug	General valve	Act normally and no leakage		
		Flipper acts up and down	Act normally		
		Spreader’s telescoping	Act normally		
		Lock and unlock	Act normally		
		Middle twistlock acts up and down	Act normally		
		Middle twistlock ‘s telescoping	Act normally		
6	Connection plate of limit device		Well fixed and no damage		
7	Power box and fast connecting plug		Well fixed and no damage		
8	Junction box on headblock		Well connected, no damage and water inleak		
The ″result″ column should be filled in :[√]pass; [0] maintain/replace; [×]rejected. This sheet should be returned after inspection finished.					

(Table 2-26)：Scheduled maintenance task list (monthly)

Equipment No.：		Start working time：	Table No.：	
Checker：		Finishing time：	Running hours：	
Maintenance personnel：		Inspection date：	Confirmed by technician：	
No.	Inspection Items	Description	Result	Solvent
Elevator				
1	Limit devices and switches	Check the function of all the emergency stop switches and limit switches.		
2	Lubrication for the elevator	Check the lubrication for all the active parts		
3	Reducers and motors	Work well and no noise; no loose bolts		
4	Control system	Work well		
5	Stanchion and its accessory	No deformation, no cracks, no loose bolts and no noises during running		
6	Brakes	Clean the brake disc, Check the abrasion of brake bush		
7	Cable	Check the twist and abnormal abrasion.		
Machinery house				
1	Cleanliness	No garbage or contamination		
2	Bolts for motors, brakes, couplings and bearing pedestal of the three main mechanisms	No looseness or deformation		
3	Wire rope clamps for the three main mechanisms	No looseness or deformation		
4	Reducers for the three main mechanisms	Proper oil level, no abnormal temperature rise, no abnormal noise and no leak.		
5	Hydraulic system of emergency brakes for the three main mechanisms	Proper oil level, no leak, correct brake torques, no looseness for pipe joints, the filter reads well.		
6	High speed brakes for the three main mechanisms	Run well; no structural damage; correct brake torque.		
7	Motors for the three main mechanisms	Run well; no abnormal temperature rise; good aeration		
8	Lighting	Work well		
9	Fan	Works well and no abnormal noise		
10	Machinery house service crane	Check worn and damaged wire ropes; check for broken wires; check the lube, Check the hook for wear, tighten the bolts; Check the cable and running condition of cable towing trolley.		
11	Electric cabinet	Check the fuse and circuit breaker; Check the contacting condition of relay and contactor; instruments read well.		
Girder				
1	Oil nipples and lube piping for anti-snag device	No leakage, distortion or rust		

2	Main structure for anti-snag device	No deformation, incomplete fusion weld or cracks.		
3	Pin shafts of anti-snag	Well lubricated, no abnormal abrasion		
4	Bolts for press plate cover of sheaves for the anti-snag	No looseness or damage		
5	Oil nipples and lubrication piping for tension device	No leakage, deformation or rust		
6	Fixing bolts for sheaves of tension device	No looseness or damage		
7	Main structure for tension device	No deformation, incomplete fusion weld or cracks.		
8	Slap blocks	Well fixed, no abnormal abrasion		
9	Outdoor service crane	Correct position for slide trolley, no loose part, no distortion and incomplete fusion weld.		
Cable chain system				
1	Cable	No damage		
2	Structure for cable towing arm	No distortion, crack or incomplete fusion weld		
3	Bolts and screws	No looseness; no screw broken.		
4	Press plates and clamps for cable	No looseness or damage		
Along the boom				
1	Rail clips and bolts for trolley	No looseness, droping or abrasion		
2	Connecting plate and locating block for short rail	No crack, incomplete fusion weld or looseness		
3	The joint of rails	No obvious concaveness, just height and horizontal deviation.		
4	Rails on boom and girder	No obvious concaveness, cracks or abnormal abrasion		
5	Furcation, guide wheel and slide rail	Well lubricated; no abrasion or looseness, correct position for guide wheel		
6	Boom operation room	No incomplete fusion weld, distortion or rain leak, check the function of the lock, the indicator lights and switches.		
7	Lubrication piping and oil nipples	No leakage, rust or distortion		
8	Boom sheaves and bearings for boom	No abrasion, distortion or looseness		
9	Supports for sheavs	No distortion, incomplete fusion weld, crack; well lubricated bearings and no abnormal abrasion for bearings		
10	Boom buffers	No aging or looseness		
11	Trolley buffers	No aging or looseness		

12	Joints shafts and the fasteners	No abnormal abrasion, distortion or looseness and well lubricated		
13	Supports for hinge points and slap blocks	No slide, no cracks and no loose bolts		
14	The Guide rollers	Rotate freely and normal abrasion		
15	Lubrication piping and oil nipples on Tie Beam at hinge point	No leakage, rust or distortion		
16	Air piping and pressure reducing vlave	No air leakage, rust or distortion; check the valve		
17	Anti-collision wire rope	Normal tension, no damage; check the function of the limit device		
18	Slap blocks under Tie Beams	No abnoraml abrasion		
19	Sheaves and fasteners at the end of trolley	No abrasion, distortion or looseness		
20	Limit device for trolley running	Well performed function, well fixed connection, no loose bolts		
21	Wire rops for trolley and Main Hoist and catenary trolley	Check for abrasion and wire broken; lubricate the wire ropes		
22	Guide Wheel for wire rope changing	No abnormal abrasion, distortion or looseness		
Boom tip				
1	Clamp plate for hoist wire rope	No looseness or distortion		
2	Trolley and catenary trolley sheaves	No looseness, distortion or abnormal abrasion		
“A” -frame platform				
1	Boom room	Clean, no sundries		
2	Connecting shafts and the copper bushes for forestay	No abnormal abrasion or noise, well lubricated		
3	Shafts and the copper bushes for top forestay	No abnormal abrasion or noise, well lubricated		
4	Boom sheaves and bearings	No abnormal abrasion, noise, distortion and looseness		
5	The rope slack limit device	Flexible motion and no distortion		
6	Protecting clamp plate and its support for rope broken	No crack, distortion or looseness		
7	Air piping and pressure reducing vlave	No air leakage, rust or distortion; check the valve		
8	Service crane	Flexible motion, no distortion or loose part		
9	Bolts of connecting plate and block plate for forestay	No distortion, incomplete fusion weld, looseness or wear		
10	Lubrication piping and	No deformation, leak or rust		

	oil nipples for top boom			
11	Top forestay and backstay and hinge base plate	No distortion, incomplete fusion weld, looseness or wear		
12	Supports for boom sheaves	No deformation, incomplete fusion weld or cracks		
13	Boom wire ropes	Check for abrasion and wire broken; lubricate the wire ropes		
14	Wind speed meter	Check the function of impeller and vane, check the connection and function for electric device.		
15	Boom limit	Well performed function, well fixed connection, no loose bolts		
16	Steel structure for boom stowage	No abrasion, incomplete fusion weld, distortion or looseness		
Operator’ s cab and trolley frame				
1	Lubrication piping and nipples for trolley	No oil leak, deformation or rust		
2	Limit devices on trolley	Well performed function, well fixed connection, no loose bolts		
3	Sheaves on trolley	No distortion, looseness or abnormal abrasion, well lubed bearings		
4	Wire rope clip plates on trolley	No distortion or looseness		
5	Guide rollers and slap blocks on the trolley	No abnormal wear, no looseness and flexible rollers		
6	Wheel supports and fasteners on trolley	No looseness, incomplete fusion weld or cracks		
7	Trolley wheels and Horizontal rollers	No rust, no abnormal abrasion; ensure the horizontal wheels do not rub the rail bolts; no loose bolts on covers.		
8	Trolley Frame	No deformation, incomplete fusion weld or cracks		
9	Trolley buffers	No distortion, leakage or damage		
10	Windows and doors for operator’ s cab	No droping or damage, lock and seal is in good condition		
11	Doors for the cab and diving platform	Open freely, no block for block plate		
12	Control desk in the cab	Check the function of pushbuttons and joysticks		
13	Guide rollers and slap blocks	No abnormal abrasion or looseness; no incomplete fusion weld on supports		
14	Cable chain towing device at trolley forehead	No distortion and looseness		
15	Cable reels on trolley	Proper oil level in gear box and no leakage; no loose bolts; no abrasion for cable; and no cracks and deformation for supports.		
Gantry				
1	Motors	Work well and no abnormal noise		

2	Reducer	Proper oil level, no abnormal temperature rise, noise or leakage.		
3	Stowage limit	Well performed function, well fixed connection, no loose bolts		
4	Dock base	No crack or damage for rail and base; clean the rail slot and stowage hole; no damage or looseness for gantry block. Check the rail: Straightness, rail guage, difference in elevation, sunken area, slope, joint misalignment and gap		
Spreader, headblock and over-height frame				
1	Main structure	No abnormal abrasion, no crack for welds		
2	Connecting bolts	No looseness		
3	Rope slack limit on headblock	Well performed function, well fixed connection, no loose bolts		
The "result" column should be filled in :[√]pass; [0] maintain/replace; [×]rejected. This sheet should be returned after inspection finished.				

(Table 2-27)：Scheduled maintenance task list (quarterly)

Equipment No.：		Start working time：	Table No.：	
Checkee：		Finishing time：	Running hours：	
Maintenance personnel：		Inspection Date:	Confirmed by technician：	
No.	Inspection Items	Description	result	Solvent
The crane				
1	Motors	No abnormal wear, fraying, chipping, or nicks, check the function of fans for motors		
2	Electric cabinet	Well performed function for switches like contactor, relay.		
3	Air conditioner	Well performed function		
4	Cables	No abnormal damage ,the megohm-meter reads well		
5	Limit devices	Well performed function, well fixed connection, no loose bolts		
6	Hydraulic systems	No leakage at pipe joints		
7	Brakes	No contamination for brake disc, no abnormal abrasion for brake bush		
8	Reducers	No abnormal noise, abnormal temperature rise or leakage, check the quality of the oil		
9	Lubrication	Lube the bearings for sheaves, wheels, bearing base and hinge; lube the pull rod pin shafts and trolley pin shafts		
10	Couplings	Check the bolts connecting and elastic parts for abrasion		
11	Drums and sheaves	Check the grooves for abrasion and corrugation		
Crane main structure, ladders and platforms				
1	Structures for ladders, platforms and railings	No incomplete fusion weld, distortion and rust		
2	Treads and fasteners	No distortion and looseness		
3	Grating plate and fastening hook for walkways	No deformation, shift or looseness		
4	Main structure	No incomplete fusion weld, distortion or cracks		
5	Painting	No rust, crimple or big area spalling		
6	High strength bolts	No looseness or rust		
7	Welds for girder and upper beam (seaside and landside)	No incomplete fusion weld or cracks		
8	Forestay and backstay	No distortion, crack; reliable function		
9	Horizontal stays	No distortion, crack; reliable fixed		
10	Portal diagonal	No distortion, crack; reliable fixed		
11	Bumping plate for limit devices	No incomplete fusion weld, distortion, rust or cracks		

12	Steel structure for gantry(equalizer beam, gantry frame, stowage support)	No distortion, welding cracking, rust or abrasion		
13	Baffle plate and anti-shear block for equalizer beam	No welding cracking or loose bolts for baffle plate		
14	Checker’ s room	No incomplete fusion weld, distortion or rain leakage, check the locks		
15	Air piping and pressure reducing valve	No gas leakage, distorion or rust; check the valve		
16	Storm protection stays and tie-downs	Flexible motion, well lubricated and fixed		
17	Air compress system	Good condition; proper oil level and no gas leakage		
18	Emergency device	Well fixed, no damage and oil leakage		
19				
20	Rope rereeving device	No abnormal noise, deformation, incomplete fusion weld, or rust		
21	Main structure for machinery house	No distortion, incomplete fusion weld and rain leakage		
22	Portal beam and rail for service crane	No abnormal abrasion, distortion and welding cracking		
23	Rail clip for trolley	No loose bolts and normal torque		
24	Hinge point	No abnormal abrasion, distortion and welding cracking		
25	Wheels for gantry and trolley	No abnormal abrasion		
26	Block for trolley	No looseness or damage		
27	Hoist motors and brakes for indoor service crane	Works well; no rust and block for brakes		
28	Rope guide and rope holding device	Work well, no slope		
29	Buffers	No aging, cracks or looseness		
30	Main structure for anti-sang device	No distortion, incomplete fusion weld or cracks		
31	Outdoor lifting crane	Correct position for slide trolley; no distortion, incomplete fusion weld or looseness for all parts.		
32	Cable chain tow arm on trolley frame	No deformation cracks or incomplete fusion weld.		
33	Hanging structure for operator’ s cab	No deformation, cracks or incomplete fusion weld		
34	Hanging bolts for operator’ s cab	No looseness, distortion or rust		
35	Steel structure for operator’ s cab	No distortion, incomplete fusion weld or rain inleakage		
36	Trolley frame	No distortion, incomplete fusion weld or cracks		

37	Supports for sheaves	No distortion, incomplete fusion weld or cracks		
38	Main structure for tension device	No distortion, incomplete fusion weld or cracks		
39	Outdoor lifting trolley	Flexible motion; no distortion and no loose parts		
40	Supports for boom sheave block	No distortion, incomplete fusion weld or cracks		
41	Steel structure for boom stowage device	No incomplete fusion weld, distortion, abrasion or looseness		
42	Headblock	No welding cracking, check the sheaves and the function of sheave bearings, well lubricated.		
The "result" column should be filled in :[√]pass; [O] maintain/replace; [×]rejected. This sheet should be returned after inspection finished.				

(Table 2-28): Scheduled maintenance task list (semi-annually)

Equipment No.:		Start time:	No. of Check sheet:	
Checker:		Finish time:	Running hours:	
Maintenance personnel:		Inspection date:	Confirmed by technican:	
No.	Inspection Items	Description	Result	Solvent
Crane and elevator				
1	Lighting	Keep reliable state for light base and rectifier, ensure the function of emergency lighting		
2	Limit devices	Check for connection and bolts		
3	Reducer for elevator	Work well, no abnormal noise, enough oil and check the quality of the oil		
4	Standard joint for elevator	No distorion or cracks, no loose bolts		
5	Elevator door, handrail, absorber and buffers	Well performed function, no distortion or abnormal abrasion		
6	The security and safety for elevator	No abnormal noise while running, correct stop, and no abnormal abrasion for guide roller;		
7	Motor and electric system for elevator	Clean the heat radiation panel for motors, check all the wires, seals, sealed pipes and connections.		
8	Alarm system, lighting and communication system for elevator	Well performed function		
Machinery house				
1	Motors for the three main mechanisms	Check the insulation condition of motors by megohm meter, clean the motor coil		
2	Electric cabinet	Check the capacitance of reactor and filter, reactance and resistance of filter, and electrical source connection of contactor and relay.		
3	Brakes	Check the function of limit devices, Check the abrasion and damage of each part		
4	Rope rereeving device	Well lubricated bearings, well connected electric equipment		
5	Bearing bases for the three main mechanisms	Check for noise, abnormal temperature rise and loose bolts, check the seal		
6	Reducers for the three main mechanisms	Well fastened, proper oil level and no leakage		
7	Service crane	Check the connection in control box		
Girder				
1	Outdoor beam crane	Lubricate the bearing		
2	Sheaves for tensioner and anti-snag device	Lubricate the bearing		
3	Trolley tensioner cylinder	Lubricate the cylinder bearing		
Cable chain system				

1	Fixing bolts and screws	No looseness or break		
2				
Operator' s cab and trolley frame				
1	Running speed and acceleration for trolley and main hoist	Good condition		
2	Slip ring box for cable reel	Clean; Check the connection of bolts, Check the interface of carbon brush for abrasion; Check the seal of box		
3	Brakes for cable reel	Good condition		
Along the boom				
1	Hinge point between boom and girder	No abnormal abrasion and noise		
2	Roller at Hinge point	No abnormal abrasion or looesness, rotate freely		
3	Trolley buffers	No distortion and damage		
4	Weight sensor	Well connected and no broken cable		
"A" -frame platform				
1	Wind speed alarm system	Well performed function and firm connection		
2	Outdoor beam crane	Lubricate the bearing, Check damage state		
Gantry				
1	Running speed and acceleration	Good condition		
2	Motors	Check the insulation condition of motors by megohm meter, clean the motor coil		
3	Couplings	Lubricate the coupling, Check the abrasion		
Spreader and headblock				
1	Oil reservior	Proper oil level and quality		
2	Filter	Function well, and reads well		
3	Twistlock	No abnormal abrasion		
4	Over-height frame	Well lubricated twistlocks and other active parts		
The "result" column should be filled in :[√]pass; [O] maintain/replace; [×]rejected. This sheet should be returned after inspection finished.				

(Table 2-29)：Scheduled maintenance task list (annualy)

Equipment No.：		Start time:	Table no.：	
Checker:		Finish time:	Running hours:	
Maintenance personnel:		Inspection date:	Confirmed by technican:	
No.	Inspection Items	Description	Result	Solvent
Crane				
1	Crane and accessory （like air compress machine）	Annual certification of service crane		
2	Electric equipment	Check the limit devices, cable, ray cable, terminal, cable number, cable groove and cable guide pipe for damage; clean the connector for ray cable		
3	Electric equipment	Check the function of loudspeaker and inter-talk communication system, chagn the power for the emergency lights		
4	Electric equipment	Make sure the fixing of all the ternimals inside the connection box, control panel, I/O equipment and control desk. No broken cables and lead		
5	Electric cabinet	Lubricate the knife switch in the high-pressure cabinet; Check the function of indicator light, switch pushbutton and gauges. No broken or exposed cable, check the resistor for over-heat or too much dust.Check the setting for over heat; over-current and time relay. Check the insulation for reactance in control cabinet.		
6	Electric parts	Checkt printing board and electric parts for abnormal state, Check the weight sensor by standard load.		
7	Transformers	Check the transformers(main transformer, lighting transformer and control transformer) to make sure no dust in the air, no dustiness and bolt connection are good for cable connector		
8	Reactance cabinet	No ablation trace for reactor, clean the parts, purge the dust. check the function of cooling fan. No break or bareness for connection		
9	Electric equipment	Keep E-room clean, fan, lights and temperature control switch in good condition		
10	Service crane	Annual certification of service crane		
11	Elevator	Annual certification of service crane , Check all the loaded bearings and other loaded parts for abrasion and rust.		
12	Air conditioner (in electric boom and operator’ s cab)	Good condition		

Machinery house				
1	Motors for the three main mechanisms	Well connected, check the function of rectifier, adjust the over-current protection of motors according to the rated current value on the nameplate		
2	Brakes	Check the abrasion and leaning abrasion of brake bush,check main structure of brakes for distortion and cracks		
3	Hydraulic system	Check the quality of hydraulic oil(e.g.hydrous rate)		
4	Drums for the three main mechanisms	Check the abrasion of drum rope groove		
5	Emergency device	Check the limit device, lubricate the chain and coupling		
6	Reducers, motors and brakes for the three main mechanisms	Check the torque of base fixing bolts , gears inside reducers and mesh of gear teeth		
7	Low speed couplings	Check the abrasion and centralization for the coupling; tighten the bolts		
8	High speed couplings	Adjust the radius and angle		
9	Service crane	Check the rope guide, safety device, control cable, control panel, brake and hanging device for wire rope stowage. Check the indoor hook for cracks, cold distortion and abnormal abrasion. Check fixing condition of base and hook.Check the function of coupling for low speed hoist motor and emergency limit switches. Tighten all the bolts.Lubricate the wire ropes		
10	The limit device on the tip of bearing base	Check the abrasion and centralization of the limit switch coupling		
Girder				
1	Main structure for anti-snag device	No welding cracking		
2	Hydraulic system for anti-snag device and tensioner	Correct pressure setting for the valves		
3	Sheaves for tensioner and anti-snag device	Check the rope groove for corrugation		
4	Cylinders for tensioner and anti-snag device	No abnormal damage or leakage		
Operator’ s cab and trolley frame				
1	Spreader cable reel	Lubricate the bearings, Check the mesh of gear teeth		
2	Trolley wheels and horizontal rollers	No abnormal abrasion		
3	Fixed guide roller	No abnormal abrasion, rotate freely; well fixed slap blocks and no welding cracking on the roller base.		

"A" -frame platform				
1	Boom wire ropes	Lube the ropes ,Check for abnormal abrasion and broken wire		Check when"boom"
Gantry cable reel				
1	Gantry cable reel	Lubricate the bearing, Check the mash of gear teeth		
2	HV slip ring box	Well sealed for the box, no water inleakage; clean the dust,well copulate slip ring and carbon brush and no sign for break; no looseness, break or bareness for connection.		
Gantry				
1	Ground junction box	No distortion or water leakage for the box,clean the dust, well connected,no damage for the wire tips		
2	Mobile high voltage cable	No distortion, break or bareness		
3	Friction bush for rail brake	Check the abrasion of brake bush		
4	Lurication	Lubricate wheel bearings and pin shaft of equalizer beam		
5	Gantry reducers	Check the quality of gear oil(e.g. hydrous rate)		
6	Hydraulic system	Check the quality of hydraulic oil(e.g. hydrous rate)		
Spreader and headblock				
1	Lubrication	Lubricate the motors and bearings		
2	Spreader, headblock and over-height frame	Check the abrasion of the lock pin		
The "result" column should be filled in :[√]pass; [O] maintain/replace; [×]rejected. This sheet should be returned after inspection finished.				

(Table 2-30)：Scheduled maintenance task list (quinquennially)

Equipment No.：		Start time：	Table no.：	
Checker：		Finish time：	Running hours：	
Maintenance personnel：		Inspection date：	Confirmed by technican：	
No.	Inspection Items	Description	Result	Solvent
Crane				
1	QC and accessory	Quinquennial certification of service crane		
2	Elevator	Carry out complete inspection on the motors, brakes and gears in reducers, replace the complete safety equipment		
Machinery house				
1	Motors for the three main mechanisms	Replace the cooling fan of the motor		
2	Reducers for the three main mechanisms	Carry out complete inspection on the gear surface and gear teeth mash.		
3	Low speed couplings for the three main mechanisms	Carry out complete inspection and heavy repair		
4	Low speed couplings for the three main mechanisms	Carry out complete inspection and heavy repair		
5	High speed brakes for the three main mechanisms	Carry out complete inspection and heavy repair		
6	Low speed brakes for the three main mechanisms	Carry out complete inspection and heavy repair		
7	Service crane	Quinquennial certification of service crane		
Operator’ s cab and trolley frame				
1	Spreader cable	Carry out complete inspection for break		
2	Reducer for spreader cable reel	Carry out complete inspection on the gear surface and gear teeth mash.		
3	Wheels for trolley	Carry out complete inspection on the wheels for wear		
4	Trolley power cables	Carry out complete inspection for break		
Gantry cable reel				
1	Reducers	Carry out complete inspection on the gear surface and gear teeth mash.		
Gantry				
1	Wheels	Carry out complete inspection on the wheels for wear		
2	Rail brakes	Carry out complete inspection and heavy repair		
3	Reducers	Carry out complete inspection on the gear surface and gear teeth mash.		
Spreader and headblock				
1	Lock pin	Carry out complete inspection on the wheels for wear and crack		
The "result" column should be filled in :[√]pass; [O] maintain/replace; [×]rejected. This sheet should be returned after inspection finished.				

(Table 2-31)：Scheduled maintenance task list (decennially)

Equipment No.：		Start time：	Table No.：	
Checker：		Finish time：	Running hours：	
Maintenance personnel：		Inspection date：	Confirmed by technican：	
No.	Inspection Items	Description	Result	Solvent
Crane				
1	QC and accessory	Quinquennial certification of service crane		
2	Elevator	Carry out complete inspection on the motors, brakes and gears in reducers, replace the complete safety equipment		
Machinery house				
1	Motors for the three main mechanisms	Carry out complete inspection and heavy repair		
2	Reducers for the three main mechanisms	Carry out complete inspection and heavy repair		
3	Low speed couplings for the three main mechanisms	Carry out complete inspection and heavy repair		
4	High speed couplings for the three main mechanisms	Carry out complete inspection and heavy repair		
5	High speed brakes for the three main mechanisms	Carry out complete inspection and heavy repair		
6	Low speed brakes for the three main mechanisms	Carry out complete inspection and heavy repair		
7	Cooler and emergency motor	Carry out complete inspection and heavy repair		
8	Main switch for high voltage cabinet	Carry out complete inspection and heavy repair		
Girder				
1	Sheaves for tensioner and anti-snag device	Carry out complete inspection or replacement		
Cable chain system				
1	Cables for chanin system	Carry out complete inspection or replacement		
Operator’ s cab and trolley frame				
1	Sheaves	Carry out complete inspection or replacement		
2	Slip ring box	Carry out complete inspection and heavy repair		
3	Spreader cables	Carry out complete inspection or replacement		
4	Trolley wheels	Carry out complete inspection or replacement		
Boom tip				
1	Trolley sheaves	Carry out complete inspection or replacement		
Gantry cable reel				
1	Cables	Carry out complete inspection or replacement		
2	High voltage slip ring box	Carry out complete inspection and heavy repair		

Gantry				
1	Reducers	Carry out complete inspection and heavy repair		
2	Rail brakes	Carry out complete inspection and heavy repair		
3	Cables	Carry out complete inspection or replacement		
Spreader and headblock				
1	Spreader and headblock	Carry out complete inspection and heavy repair		
The "result" column should be filled in :[√]pass; [O] maintain/replace; [×]rejected. This sheet should be returned after inspection finished.				